

Module 4: Modern and Renewable Energy sources

Syllabus: Nuclear fission and fusion; Different Types of Nuclear Reactors-Thermal/Fast Neutron-Description and working of Gas Cooled, Boiling Water (BWR), Pressurised Water (PWR), Pressurised Heavy Water (PHWR) and Fast Breeder Reactor (FBR) Nuclear Reactors. Solar Cells: Solar spectrum, photovoltaic effect, structure and working principle of solar cell, I-V characteristics, power conversion efficiency, materials for PV, emerging PV technologies for alternative energy devices. Introduction to other Renewable Energy sources-Wind-Tidal-Geothermal systems

4.1 Modern and Renewable Energy sources

Renewable energy sources derive their energy from existing flows of energy from ongoing natural processes, such as sunshine, wind, flowing water, biological processes, and geothermal heat flows. A general definition of renewable energy sources is that renewable energy is captured from an energy resource that is replaced rapidly by a natural process such as power generated from the sun or from the wind. Currently, the most promising (aka economically most feasible) alternative energy sources include wind power, solar power, and hydroelectric power. Other renewable sources include geothermal and ocean energies, as well as biomass and ethanol as renewable fuels.

Modern and renewable energy sources play a crucial role in reducing carbon emissions and promoting a sustainable future. Here are some key characteristics of nuclear, solar, wind, tidal, and geothermal systems:

1. Nuclear Energy: Nuclear power plants generate electricity through the process of nuclear fission, where atoms are split to release large amounts of energy. It is a low-carbon option that offers a consistent and reliable power supply. However, concerns about radioactive waste disposal and the potential for accidents make it a controversial choice.

2. Solar Energy: Solar power harnesses the energy from the sun through photovoltaic (PV) cells or solar thermal collectors. It is a clean and abundant source of energy, with no emissions or pollution during operation. Solar panels can be installed on rooftops or in solar farms, making it suitable for both residential and commercial use.

3. Wind Energy: Wind turbines convert wind energy into electricity through the rotation of their blades. Wind power is a clean and sustainable energy source that produces no greenhouse gas emissions. It can be harnessed both onshore and offshore, with larger turbines capable of generating significant amounts of electricity.

4. Tidal Energy: Tidal power utilizes the energy from the gravitational pull of the moon and sun on the Earth's tides. Tidal barrages or tidal turbines capture the kinetic energy of moving water and convert it into electricity. Tidal energy is predictable, reliable, and renewable, but it is limited to coastal regions with significant tidal ranges.

5. Geothermal Energy: Geothermal power taps into the heat stored beneath the Earth's surface. It involves extracting hot water or steam to generate electricity or provide heating and cooling. Geothermal energy is renewable, reliable, and emits minimal greenhouse gases. However, it is geographically constrained to areas with accessible geothermal reservoirs.

Each of these modern and renewable energy sources has its advantages and considerations. The choice of energy system depends on factors such as location, available resources, cost, and environmental impact. The transition to a mix of these energy sources is vital for a sustainable and cleaner energy future.

4.2 Nuclear Energy

The nucleus of an atom occupies an astonishingly small volume at the center of the atom and is most densely packed with protons and neutrons. Almost all the entire mass of the atom is accounted by the nucleus alone. The particles in the nucleus are held together by nuclear forces which are a fundamental force like gravitational force. The heavier nuclei exhibit

instability and transform into stable nuclei through radioactive disintegration. The natural radioactivity of nuclei is effectively used in the determination of ages of organic as well as inorganic substances. Radioactivity can be induced in nuclei through nuclear reactions, which assist mankind in different fields such as medicine, industry and agriculture. The neutron induced nuclear reactions occupy special place. The neutrons have the ability of causing fission of uranium and plutonium nuclei which liberates tremendous energy. The nucleus is thus a storehouse of enormous power, which can be utilized in wars as well as in production of electrical power. For the past century fossil fuels namely, coal, oil and natural gas have supplied the major portion of our energy requirements. These sources will be nearly exhausted in the near future and it is imminent that alternative sources of power are to be searched for. Nuclear power is one of the alternative sources. The energy reserve in the form of uranium is many times greater than that of fossil fuels. The generation of power through nuclear fission, however, poses its own threats and problems. Nuclear fusion power is believed to be an inexhaustible source of energy and is without any attendant hazards. Efforts are in progress for producing electrical power through nuclear fusion on commercial scale.

4.2.1 The atomic nucleus The British physicist Ernest Rutherford proposed the nuclear model of atom in 1911 in an attempt to explain the large angle scattering of α -particles which were directed at a thin gold foil. He postulated that all the positive charges of an atom were concentrated in a central massive core and named it nucleus. A **nucleus** is the centre of the atom where most of the mass of the atom and all of the positive charge are concentrated.

There is a vast difference between the size of an atom and its nucleus. The nucleus is a very small part of an atom. The radius of nucleus is of the order of 10^{-14} m while that of an atom is of the order of a few angstroms (10^{-10} m). Therefore, an atom occupies about a million times more space than does a nucleus. In between the atomic electrons and the nucleus, there is a lot of void of space.

An atomic nucleus is not a single indivisible point mass. It is composed of smaller particles. It is obvious that nucleus contains protons. However, all of the nuclear mass could not be accounted for by protons. In 1920, Rutherford suggested the existence of a neutral particle in the nucleus which he called a **neutron**. Neutron was discovered in 1932 by James Chadwick. All nuclei, with the exception of hydrogen nucleus, contain neutrons. Protons and neutrons are about 2000 times more massive than electrons. They are collectively known as nucleons. Since a stable atom is electrically neutral, the number of positively charged protons in the nucleus is always equal to the number of negatively charged electrons around the nucleus. The number of protons or electrons in an atom is known as the **atomic number** and is denoted by Z . Total number of protons and neutrons in an atomic nucleus is called the **mass number** and is denoted by A . A special notation is used to designate a particular nucleus. The nucleus of an atom is denoted by the chemical symbol of the atom subscripted and superscripted respectively by atomic number Z and mass number A . Thus, if the chemical of an atom is X , its nucleus is denoted by A_ZX . For example, ${}^{12}_6C$ (read as carbon-six-twelve) denotes the nucleus of carbon atom, containing 6 protons and $12 - 6 = 6$ neutrons. There are 92 stable species of atom available in nature.

4.2.2 ISOTOPES

A particular kind of atom of any element is called a **nuclide**. A nuclide is distinguished from other nuclides by the number of protons and neutrons it contains. The atomic number Z determines the chemical nature of an element. Although for a particular element the number of electrons and protons is fixed, the number of neutrons in the nucleus may vary. It implies that the mass number A may differ though the atomic number Z remains the same. Such atoms will be chemically identical but their nuclei show marked differences in stability. Nuclei of the same element having different numbers of neutrons are called isotopes. Thus, **isotopes** are atoms of a given element that have different masses. To cite an example,

hydrogen has three isotopes. H is the most common isotope. It is just a proton. The other isotopes 2H and 3H are called deuterium and tritium respectively. In atomic form they are known as deuterium and tritium respectively. The deuterium denoted by D is made of one proton and one neutron. For about every 6500 atoms of ordinary hydrogen in water, there is one atom of deuterium. Tritium denoted by T is made of one proton and two neutrons. It is radioactive with a half-life period of 12.26 years.

The uranium element which plays a very important role in the production of nuclear energy exists in three isotopic forms.

Isotope	Relative abundance	Half-life
$^{238}_{92}U$	99.28%	4.5×10^9 years
$^{235}_{92}U$	0.714%	7.1×10^8 years
$^{234}_{92}U$	0.006%	2.5×10^5 years

4.2.3 THE NUCLEAR FORCE

The nucleons are clustered together within the small volume of the nucleus. Large repulsive electrical forces operate between the positively charged protons, which tend to tear the nucleus apart. Yet the nuclei of most atoms are stable. It means that there must be some strong attractive force which more than balances the electrostatic repulsion and holds the nucleus together. This strong attractive force is called the nuclear force. The nuclear interaction between nucleons is, therefore, a strong interaction. The nuclear force is a fundamental force like the gravitational and electrical forces, but is more complicated and not completely understood. The nuclear forces have the following properties.

Salient Features of the Nuclear Forces:

1. The nuclear force is strongly attractive and much larger in magnitude than either the electrostatic force or gravitational force. They act between any pair of nucleons – proton-proton, proton-neutron, and neutron-neutron.

2. Nuclear forces are very **short-ranged** forces. A nucleon interacts only with its nearest neighbours, over distances of the order of 10^{-15} m.
3. Nuclear forces are **charge-independent**. They do not depend on the charge of nucleons. The nuclear forces acting between two protons, between a proton and a neutron or between two neutrons have the same magnitude. That is n-n force is no different from the p-p and n-p force.
4. Nuclear forces are **spin-dependent** and depend on the mutual orientation of the spins of the nucleons. For example, a neutron and a proton are kept together forming a nucleus of deuteron, only if their spins are parallel to each other.
5. Nuclear forces are **not central** forces. They cannot be represented as directed along the straight line connecting the centers of the interacting nucleons.
6. Nuclear forces have the property of saturation. It means that each nucleon in a nucleus interacts with a limited number of neighbours. Saturation manifests itself in that the binding energy per nucleon does not grow with an increase in the number of nucleons, but remains approximately constant. Further, the saturation of the nuclear forces is indicated by the volume of a nucleus being proportional to the number of nucleons forming it.
7. The magnitude of nuclear force is so high that the work required to divide a nucleus into its constituents is about 8 MeV in contrast to a few eV required to separate the extra-nuclear electrons from its atom.

4.2.3.1 Proton neutron Theory

For elements of low mass numbers, the atomic number Z is nearly half the mass number A . Thus the number of protons is nearly equal to the number of neutrons. With increasing mass number, the value of Z becomes less than half of A which means that the number of neutrons exceeds that of protons. In case of ${}^{238}_{92}\text{U}$, $Z=92$ and $A = 238$. Therefore, $(A-Z) = 146$. Thus, uranium nucleus consists of 92 protons and 146 neutrons. For elements of low mass numbers,

the neutron-to-proton ratio is close to unity. But as the number of protons in a nucleus exceeds 20, the neutron to proton ratio is greater than unity in stable nuclides. It appears that in order to maintain nuclear stability, the neutrons must exceed protons in number (**Fig. 4.1**). Further, the neutron excess increases with the increasing atomic number. For the heaviest stable nuclides such as $^{208}_{82}\text{Pb}$ the neutron-to-proton ratio exceeds 1.5. It is inferred from the above facts that neutrons act as a nuclear “glue” to hold the nucleus together. As the number of protons increases for larger stable nucleus, the number of neutrons also increases, so that the short range nuclear forces are greater than the long range repulsive electrical forces. Neutrons provide attractive nuclear forces between both protons and other neutrons. The proton-neutron theory was put forward by Heisenberg in 1932.

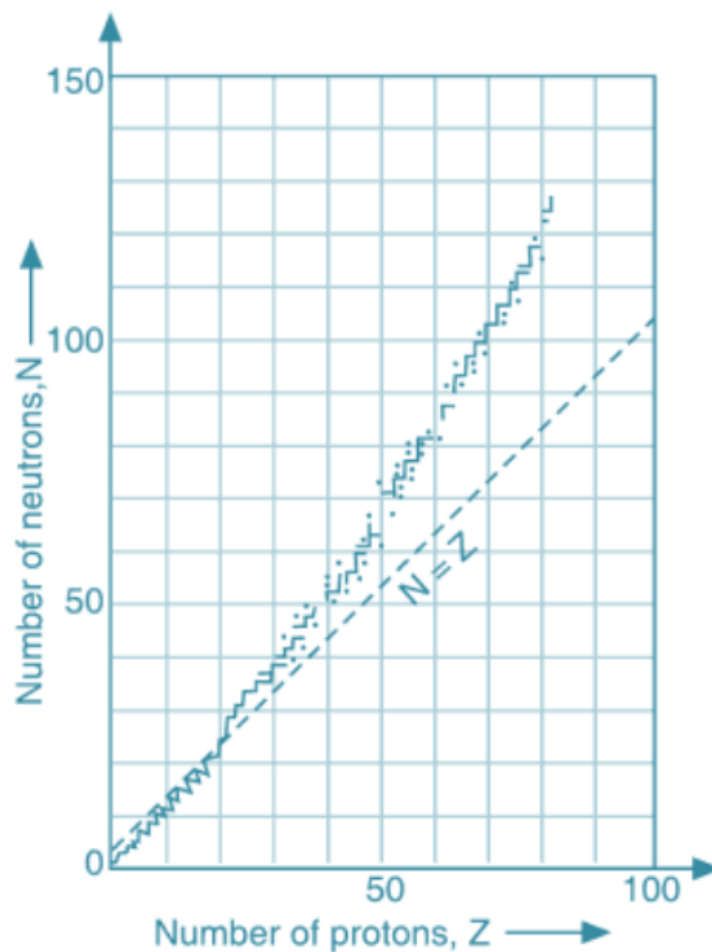


Fig. 4.1: A Plot of neutron number N versus proton number Z in nuclides. Nuclides with proton numbers greater than 20 have more neutrons than protons. The dashed line corresponding to the condition $N = Z$ is called the line of stability.

4.2.4 STATIC PROPERTIES OF NUCLEUS

4.2.4.1 Nuclear Mass

Mass of an atom, a nucleus or an elementary particle is extremely small. Expressing their masses in conventional units involves cumbersome negative powers of ten. Therefore, a separate unit called atomic mass unit has been devised for expressing the masses of atomic particles. Nuclear mass means the mass or weight of the nucleus alone. Atomic mass unit (a.m.u.) is defined as 1/12th the mass of the $^{12}_6\text{C}$ atom.

$$1 \text{ a.m.u.} = 1/12 \text{ Mass of one carbon atom} = 1/12 \times 12 \text{ kg} / 6.02 \times 10^{26} = 1.66 \times 10^{-27} \text{ kg.}$$

The uranium nucleus U-238 has a mass of 238 a.m.u. As the mass of the electron is negligibly small in comparison to that of protons and neutrons, the mass of the nucleus is taken as the mass of the corresponding atom also. Thus, the mass of the uranium atom is 238 a.m.u.

The a.m.u. is now called unified atomic mass and is denoted by the letter “u”.

4.2.4.2 Nuclear Radius

Nuclear radii are estimated from the measurements on the maximum scattering angles of α -particles when they approached the target nuclei. The nucleus is assumed to be spherical and the following empirical relation gives its radius.

$$R = R_0 A^{\frac{1}{3}}$$

where R_0 has an average value of $1.4 \times 10^{-15} \text{ m}$.

The radius of uranium nucleus = $1.4 \times 10^{-15} \text{ m} \times 238^{1/3} = 8.68 \times 10^{-15} \text{ m}$.

4.2.4.3 Nuclear Density

The density of nuclear matter is tremendous. All nuclei have nearly the same density. The density is calculated as follows.

$$\rho = \frac{\text{Mass of nucleus}}{\text{Volume of the nucleus}} = \frac{M_N}{\frac{4}{3}\pi R^3} = \frac{m_p A}{\frac{4}{3}\pi R^3}$$

$$= \frac{1.673 \times 10^{-27} \text{ kg}}{11.5 \times 10^{-45} \text{ m}^3} = 1.45 \times 10^{17} \text{ kg/m}^3$$

4.2.4.4 Nuclear Charge

Nucleus is electrically positive and the magnitude of its charge is equal to the number of protons in the nucleus. Thus, the charge on a nucleus is equal to its atomic number, Z. Thus, a uranium nucleus carries a charge of 92 units.

4.2.4.5 Nuclear Quantum States

The studies of α - and β -ray spectra show that every nucleus possesses a set of quantum states. Transitions between different nuclear states cause emission of γ -rays.

4.2.4.6 Spin and Magnetic Moment

The hyperfine structures observed in atomic spectra indicated that the nucleus has spin motion. It is concluded on the basis of experimental evidence that the protons and neutrons are in continuous motion in discrete quantized orbits. Because of this motion, the nucleus possesses angular momentum and magnetic moment.

The magnetic moment of nuclei is given by

$$\mu_1 = g \frac{h}{2\pi} \cdot \frac{e}{2M}$$

The product $\frac{h}{2\pi} \cdot \frac{e}{2M}$, often written as $eh/2M$ is known as **nuclear magneton**.

4.2.5 MASS DEFECT

The nucleus is formed by bringing protons and neutrons together. The mass of the resulting nucleus is less than the sum of the masses of the constituent protons and neutrons. This mass difference is called mass defect and is denoted by Δm .

If Z is the number of protons in the nucleus, then the number of neutrons in the nucleus is (A-Z). If m_p is the mass of the proton and m_n is that of the neutron, then the sum of the masses of the protons and neutrons

$$= Zm_p + (A - Z)m_n$$

If M is the actual mass of the nucleus, then mass defect is $\Delta m = Zm_p + (A - Z)m_n - M$

Because the nucleus is more stable than the separated neutrons and protons, the nucleus is in a lower energy state. It implies that the mass, which disappears, is released in the form of energy when nucleons are bound together in a nucleus.

For example, a ${}^4_2\text{He}$ nucleus is formed from two neutrons and two protons.

Mass of two protons $= 2 \times 1.007826 = 2.015652 \text{ amu}$

Mass of two neutrons $= 2 \times 1.008665 = 2.017330 \text{ amu}$

Total mass $= 4.032982 \text{ amu}$

Measured mass of ${}^4_2\text{He} = 4.002604 \text{ amu}$

Mass defect $\Delta m = (4.032982 - 4.002604) \text{ amu} = 0.030378 \text{ amu}$

Atoms with atomic numbers between 30 and 63 have a greater mass defect per nuclear particle than very light elements or very heavy ones, as seen in Fig. 4.2. The most stable nuclei are in the atomic number range from 30 to 63.

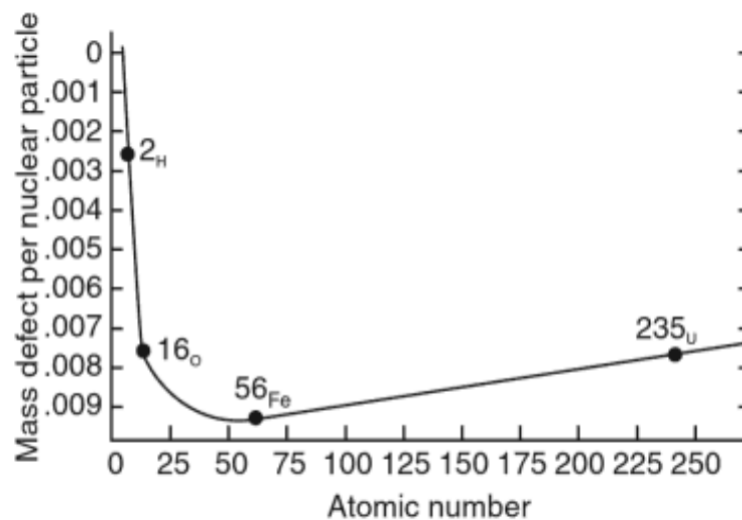


Fig. 4.2: Mass defect for different nuclides. The most stable nuclei center around Fe-56 which has the largest mass defect per nucleon.

4.2.6 BINDING ENERGY

The energy required to remove any nucleon from the nucleus is called the binding energy of that nucleon in the nucleus. It would be equal to the work that must be done in order to

remove the nucleon from the nucleus without imparting it any kinetic energy. The total binding energy of a nucleus is defined as the energy required to break up the nucleus into its constituent protons and neutrons and place them at rest at infinite distances from one another. It means that the binding energy is the energy equivalent of the mass defect. Thus,

$$\Delta E_b = (\Delta m)c^2$$

$$\Delta E_b = [Zm_p + (A - Z)m_n - M]c^2$$

The binding energy of the nucleus in MeV is expressed

NUCLEAR ENERGY

Energy requirements of mankind are at present met by fossil fuels, namely coal, oil and natural gas. These sources are being fast depleted and will be nearly exhausted in the near future. It is necessary that alternative sources of power are to be searched for. Nuclear power is one of the alternative sources. The energy reserve in the form of uranium is many times greater than that of fossil fuels. The nucleus is a store-house of enormous power. Heavier nuclei, uranium and plutonium, can be split into fragments by neutrons and such a fission process liberates tremendous energy. Nuclear fusion power is believed to be an inexhaustible source of energy and is without the radioactive hazards and pollution threats.

NUCLEAR REACTORS

We can sum up the three important features of nuclear fission reaction as follows:

(a) Energy Emission: Each fission reaction produces about 200 MeV of energy and the total energy produced by a small sample of uranium, say 1 gm, is as enormous as $8.2 \times 10^{10} \text{ J} = 22.8 \text{ MWh}$. Most of the energy is carried by fission fragments in the form of kinetic energy.

(b) Neutron Multiplicity: Each fission event is accompanied by the emission of a number of neutrons which in turn cause fission of other nuclei.

(c) Delayed Neutrons: Many of the neutrons are emitted at the instant of fission. They are known as prompt neutrons. About 1% of the neutrons are emitted due to decays of fission fragments and they are called delayed neutrons.

The last feature enables mechanical control of the reaction rate and keeps the reaction from proceeding too rapidly. The second feature can be used to build a self-sustaining chain of nuclear fissions. The energy produced in the reactions can be extracted as heat and used to boil water, the resulting steam can then be used to drive a turbine to generate electrical power.

A system in which nuclear fission is produced in a controlled, self-sustaining chain reaction is known as nuclear reactor. Enrico Fermi built the first nuclear reactor in 1942. A very large number of nuclear reactors are in operation in different countries as on today. They differ widely in design and construction depending on their use. A nuclear reactor essentially consists of seven components arranged in different zones, each of which serves a definite purpose.

Fuel

Nuclear reactors can use pure fissile materials but it is easier and cheaper to use mixtures of isotopes. Naturally occurring uranium consists of 99.3% U-238 and only 0.7% U-235. U-238 is for all practical purposes not fissionable. Often natural uranium in which there is one U-235 atom per 140 U-238 atoms is used as nuclear fuel. In order to be useful as a fuel, the concentration of U-235 must be substantially increased (upto 3%) in naturally available uranium. This process is called enrichment. In many reactors, uranium enriched in U-235 is used as fuel. Another fissionable material is Pu-239. This does not occur in nature. It is produced through neutron capture by the non-fissionable U-238. The process of plutonium fuel production from uranium is known as breeding.

Moderators

The kinetic energy of neutrons emitted in fission process is of the order of a few MeV. Such high energy neutrons are known as fast neutrons. The fast neutrons have relatively low probability of inducing fission. Nuclear fission is much more probable with slow neutrons. The fast neutrons must be therefore, slowed down in order to increase their chances of initiating fission events. If a neutron is scattered from a heavy nucleus like uranium, the energy of the neutron is not changed. On the other hand in a collision with a very light nucleus, the neutron can lose substantial energy. Materials consisting of atoms of lower atomic mass are used for slowing down neutrons. Such materials are called moderators. The

commonly used moderators are water, heavy water, graphite and beryllium. Heavy water is considered to be the best moderator, since it does not absorb neutrons. When a solid moderator is required, graphite is commonly used. Ordinary water is a good moderator, but has high neutron absorption cross-section. Beryllium and its oxide are also good moderators, but they are expensive, toxic and have poor mechanical property.

Reflectors

Some of the neutrons generated in the fission process may leak away without being absorbed. It is necessary to conserve neutrons so that we minimize the consumption of fissile material and keep the size of the reactors small. To reduce the neutron loss due to leakage the inner surface of the reactor is surrounded by a material which reflects the neutrons back into the core. Such materials are called reflectors. Reflectors are made of nickel, thorium or other suitable materials.

Coolants

Intense heat is generated within the reactor core due to nuclear fission reactions. This heat must be removed using coolants for the safe operation of reactor. Ordinary water, heavy water and liquid metals are used as coolants. The coolant keeps the fuel assembly at a safe temperature; at the same time the heat carried away by the coolant is used in the heat exchange for utilization in power generation.

Control Rods

The rate of reactions in a nuclear reactor is controlled by control rods. Since the neutrons are responsible for the progress of the chain reactions, suitable neutron absorbers are employed to achieve control of reaction rate. Cadmium and boron are the most frequently used materials. The control procedure involves the insertion or withdrawal of these materials, taken in the form of rods, into or from the reactor core. With the control rod fully inserted, enough neutrons are absorbed so that the average number of neutrons available to cause new fissions

is less than one per fission reaction. As the rod is slowly withdrawn, the average number of available neutrons increases until it is just equal to one per reaction. At that time the reactor is said to be critical. During the operation, the position of the control rod is continually adjusted so that energy is released at a steady rate.

Structural and Cladding materials

In a reactor, structural materials have to be used in the form of mechanical support for the various components. They are also used for holding the fuel, coolants, control rods and measuring instruments. Uranium readily reacts with air, water and other fluids. Hence cladding is required to pack fuel elements to isolate them. The cladding also prevents escape of the fission fragments. Zirconium has been found to be an excellent structural and cladding material. Titanium is also a good material but it is very expensive.

Reactor Shielding

A nuclear reactor is a powerful source of highly penetrating neutrons and γ -radiation, which are very harmful to life. Therefore, a thermal shield and a biological shield are used to minimize the effects of the harmful radiation. The thermal shield, usually a wall of steel, is placed between the reactor and the biological shield to protect the latter from excessive heating and damage. The biological shield is usually made of concrete. Concrete wall of about 2 m thick surrounding the reactor serves as an adequate shield.

Nuclear Power

- * It uses nuclear energy to generate power.
- * Nuclear fuels used in power industry are principally uranium and thorium.
- * One kg of Uranium can produce the energy equivalent to burning of about 4×10^6 kg of High grade coal.

Advantages:

- * Nuclear Power plants high initial cost but has low operating cost compared to conventional thermal power plants.
- * These plants become more economical if built in large capacities.
- * It reduces the demand on depleting resources of energy (coal, oil and gas)
- * The problem of transportation of conventional fuels is avoided since requirement of nuclear fuel for nuclear power station is negligible.
- * Large storage facilities of fuel are not needed.

Principles of nuclear energy:

Nuclear Binding Energy:

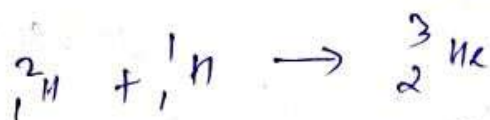
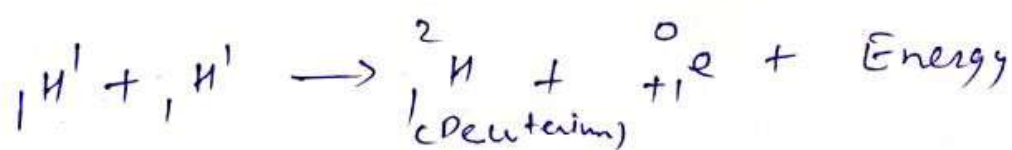
Nucleus is made up of protons and neutrons, but the mass of a nucleus is always less than the sum of the individual masses of protons and neutrons.

Nuclear Fusion!

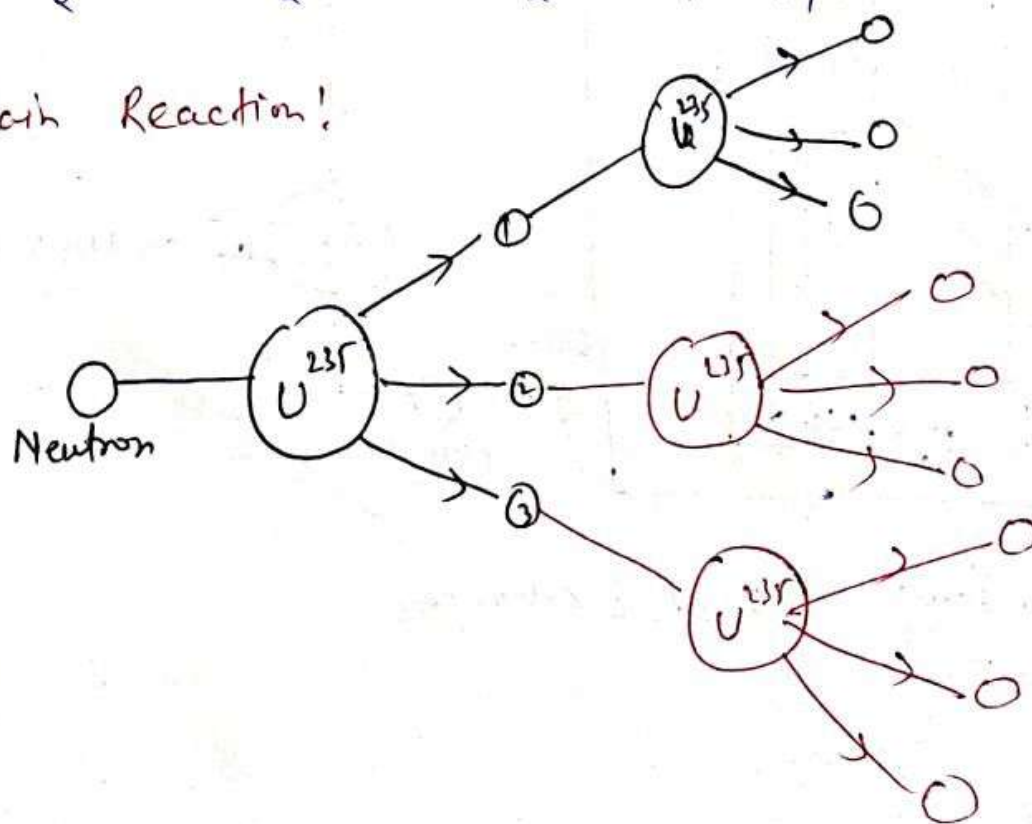
* Nuclear fusion is the process in which two lighter nuclei are combined to be fused together to form a heavier and stable nucleus.

* The mass of the product nucleus formed is less than the sum of the masses of the nuclei fused.

* This mass defect is converted into energy.



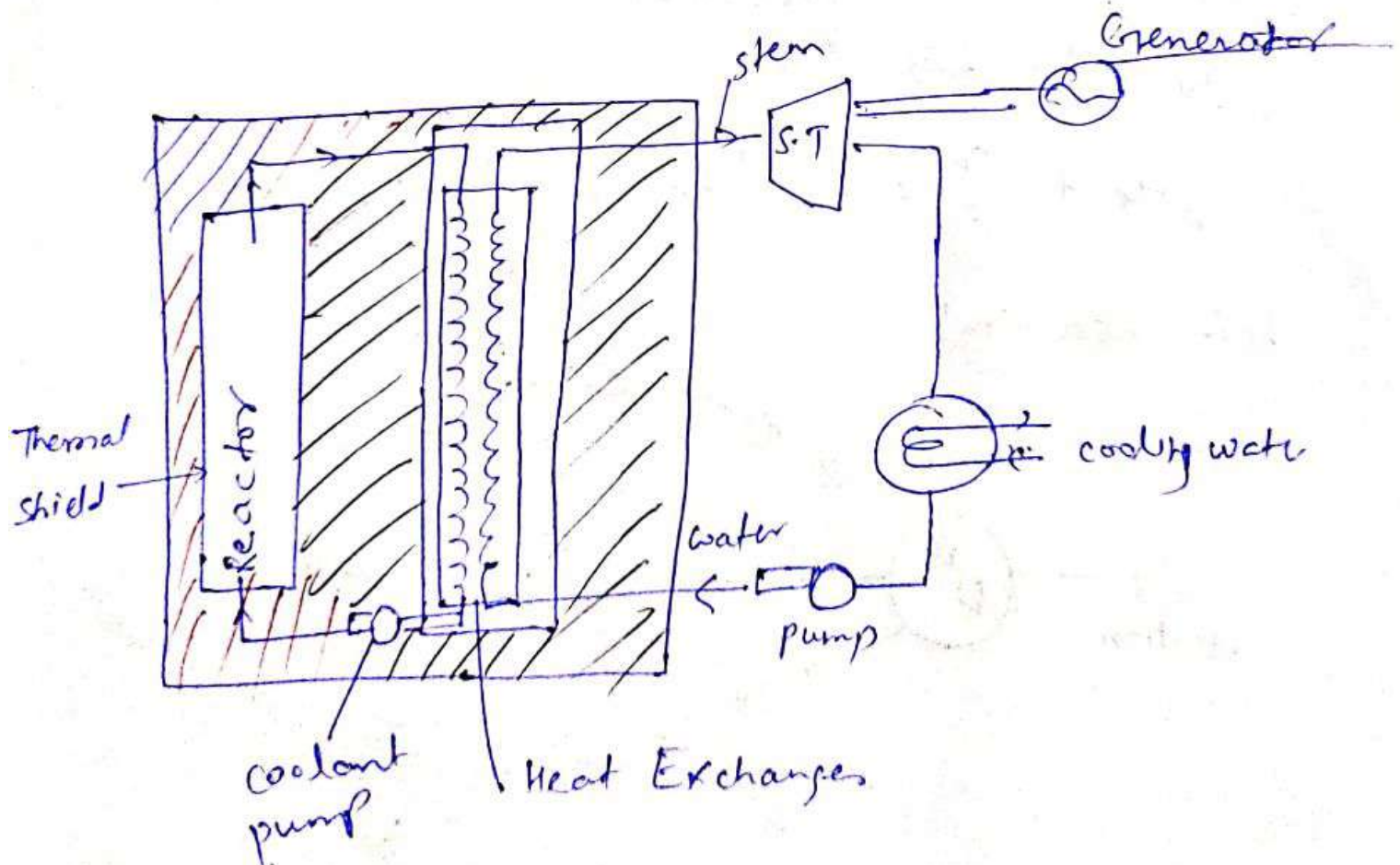
Chain Reaction!



Elements of Nuclear Power Plant:

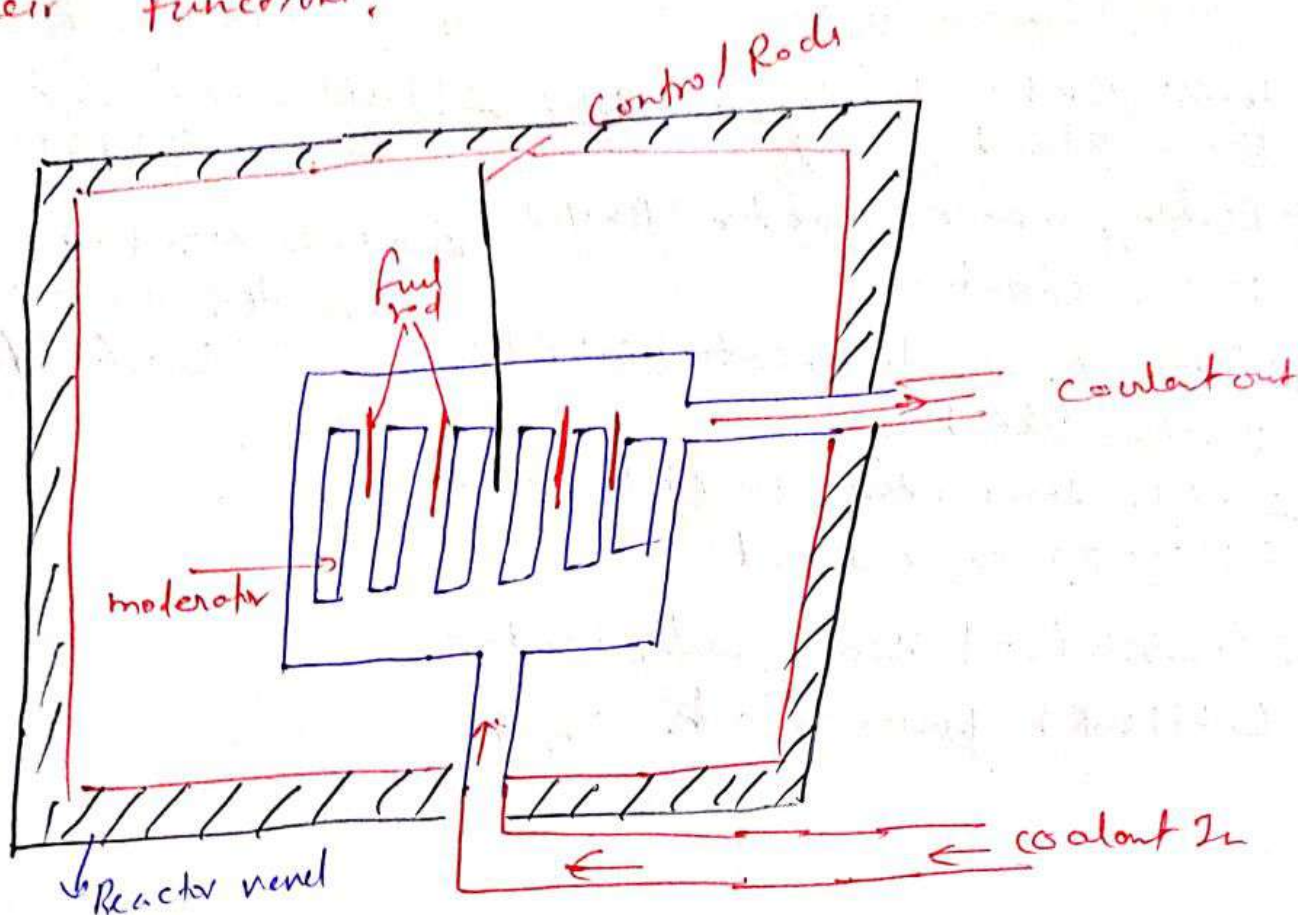
(4)

- * Nuclear Reactor
- * Heat exchanger
- * Coolant and coolant pump
- * Steam turbine, condenser and generator.
- * Nuclear Reactor is used to generate heat energy
- * This heat energy is used in a heat exchanger for formation of steam which is used for power generation.
- * Nuclear Power plants are used as base load plants



Main Component of Nuclear Reactor and Their functions!

5



Fuel Rod: ${}_{92}^{235}\text{U}$, ${}_{94}^{239}\text{Pu}$ and ${}_{92}^{233}\text{U}$

* The fuel rod is fabricated in various shapes like rods, plates, pins, pellets etc.

Moderator: * The moderator are the substance which help in reducing the speed of neutrons. and
* moderators are light water (H_2O), deuterium (D_2O), graphite and beryllium.

Control Rods: * It is used to control the chain reaction by absorbing required neutrons.

* materials used are boron and cadmium.

Reflector: * It is desirable that the neutrons are not allowed to escape the reactor core. Hence a reflector surrounding is provided.

Types of Nuclear Reactor Power Plants!

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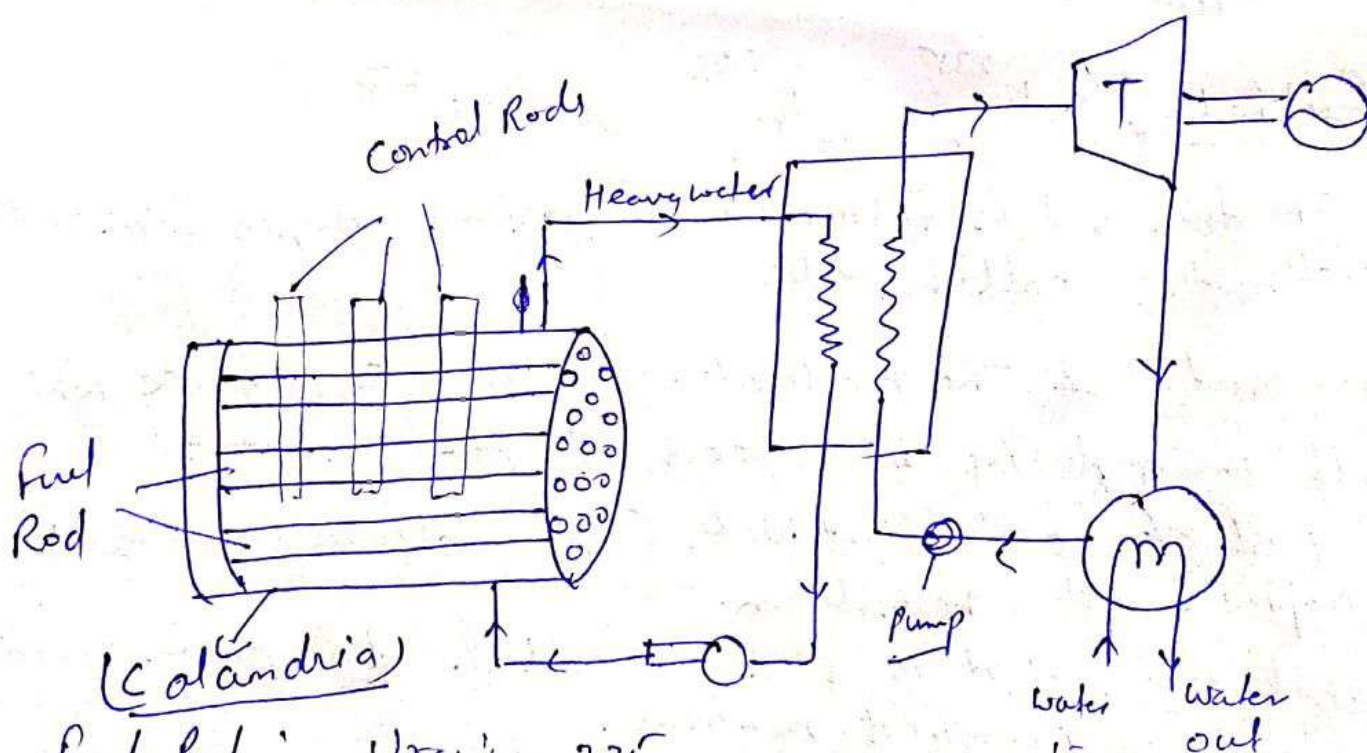
Thermal Reactors

- * Pressurized water Reactor (PWR) power plant (Imp.)
- * Boiling water Reactor (BWR) power plant
- * The gas cooled Reactor (GCR) power plant
- * High temperature gas cooled reactor (HTGR) power plant
- * Pressurized Heavy water Reactor (PHWR) power plants (Imp.)

Fast Breeder Reactor

- * Liquid metal fast breeder Reactor (LMFBR)
- * Gas cooled fast breeder reactor (GCFBR)

CANDU Reactor (Canadian Deuterium Uranium)



Fuel Rod :- Uranium 235

coolant and moderator = D_2O

10.15 The Pressurized Water Reactor (PWR) Power Plant :

The schematic diagram of PWR power plant is shown in Fig. 10.15.1.

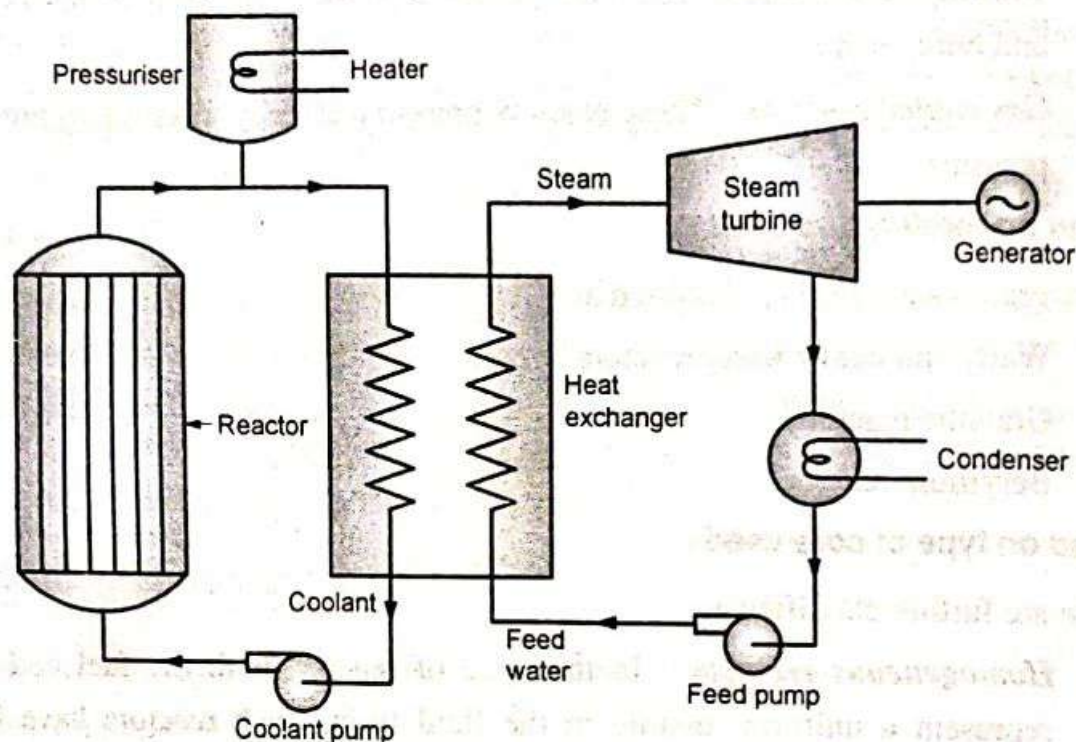


Fig. 10.15.1 : Pressurized water reactor power plant

It uses enriched uranium. The water or heavy water under pressure is used both as coolant and moderator. Before starting the reactor the water in pressuriser is boiled and converted into steam by electric heating coil. In order to prevent the boiling of water in the core, it is kept under pressure of about 130 to 150 bar. It helps in absorbing the heat by water in the liquid state in the reactor.

The heat energy absorbed by water in the reactor is used for converting the water into steam in a heat exchanger. This steam is used in a conventional way in the steam power plant cycle as shown in Fig. 10.15.1. The water coolant from heat exchanger is recirculated to the reactor with the help of coolant pump.

These power plants are compact and its cost is reduced since it uses water both as coolant and moderator. However, the high pressure in the primary circuit of water absorbing heat in the reactor requires a strong reactor shell which increases its cost.

In this reactors, the water flowing through the reactor becomes radioactive, therefore this primary circuit must be heavily shielded to protect the operators.

10.15.1 Advantages of PWR :

1. Water is used both as coolant and moderator which is cheap and easily available.
2. Reactor is compact.
3. Small number of control rods are required.
4. Fission products remain contained in the reactor.

10.15.2 Disadvantages of PWR :

1. Capital cost is high since the reactor and primary circuit works under pressure.
2. Only saturated steam can be generated in secondary circuit, therefore the efficiency of plant is low.
3. Costly shielding is required to shield the operators in primary circuit since the coolant becomes radioactive.
4. Severe corrosion problems.
5. Fuel suffers from radiations. Therefore, its reprocessing is difficult.
6. Plants need to be shut down for fuel charging.

10.16 The Boiling Water Reactor (BWR) Power Plant :

The arrangement of a boiling water reactor power plant is shown in Fig. 10.16.1.

It uses **enriched uranium** as fuel and water is used both as coolant and moderator. These reactors do not require a heat exchanger as needed in case of pressurized water reactors since the water is directly converted into saturated steam at about 285°C at 70 bar pressure.

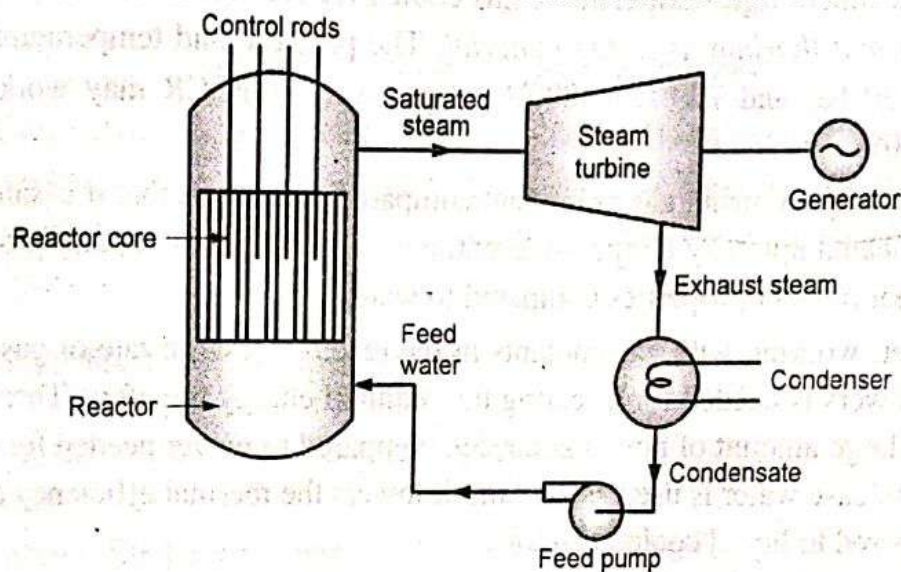


Fig. 10.16.1 : Boiling water reactor

For these reason, this system is also called as direct cycle boiling water reactor power plant.

The feed water circulated in the reactor is converted into saturated steam by transfer of energy released by fission in the reactor core. This steam is supplied to steam turbine in a conventional power plant working on the cycle. The mechanical power developed by the turbine is converted into electric energy by the generator. The exhaust of steam from turbine is condensed in the condenser. The condensate is returned to reactor as feed water by the feed water pump.

In India the Tarapur atomic power plant works on BWR principle.

10.16.1 Advantages of BWR :

1. It eliminates the use of heat exchanger, pressuriser, circulating pump and piping. Therefore system is simple, cheap and efficiency is high.
2. Use of low pressure reactor further reduces the cost of plant.

10.16.2 Disadvantages of BWR :

1. It has the possibility of radioactive contamination of steam turbine.
2. It cannot meet the sudden changes in load on the plant.
3. System requires extensive safety devices against radioactive radiations which are costly.

10.17 The Gas Cooled Reactors (GCR) :

A gas cooled reactor was first developed in U.K. uses CO_2 as coolant instead of water and graphite as moderator. It is called *gas cooled graphite moderated (GCGM) reactor*. It uses natural uranium as fuel. The coolant pressure is about 7 bar and temperature $336^\circ C$.

Another gas cooled reactor developed in U.S.A uses helium as coolant and graphite as moderator. It is called **high temperature gas cooled (HTGC)** reactor. The fuel used in U^{233} as fissile material and thorium as fertile material. The pressure and temperatures of the coolant are 15 bar to 30 bar and $700^\circ C$ to $800^\circ C$ respectively. HTGCR may work upto a thermal efficiency of 40%.

The advantage of using gas as coolant compared to water is that it is safe, easy to handle and it can be heated upto any temperature without change of phase at any pressure though the gas has low heat transfer properties compared to water.

However, working with gas coolants in the reactors, a large rate of gas circulation with the help of blowers is needed for affecting the required energy transfers. Therefore for driving the blowers a large amount of power is needed compared to power needed for running the feed water pumps in case water is used as coolant. It lowers the thermal efficiency of the gas cooled reactors compared to liquid cooled reactors.

1. Advantages of gas cooled reactor :

1. It has no corrosion problem.
2. Gases are safe and easy to handle.
3. Graphite remains stable at high temperatures and radiation problems are minimum.
4. These can be operated at high temperatures.
5. Gases can be pressurized easily.

2. Disadvantages of gas cooled reactor :

1. Gases have lower heat transfer coefficient thus it requires large heat exchangers.
2. Fuels have to be operated at high temperatures.

3. Large amount of fuel loading is required.
 4. If helium is used as in case of HTGC, leakage is a major problem.
 5. More power is needed for coolant circulation compared to liquid cooled reactors.
- Schematic diagram of gas cooled reactor power plant is shown in Fig. 10.17.1.

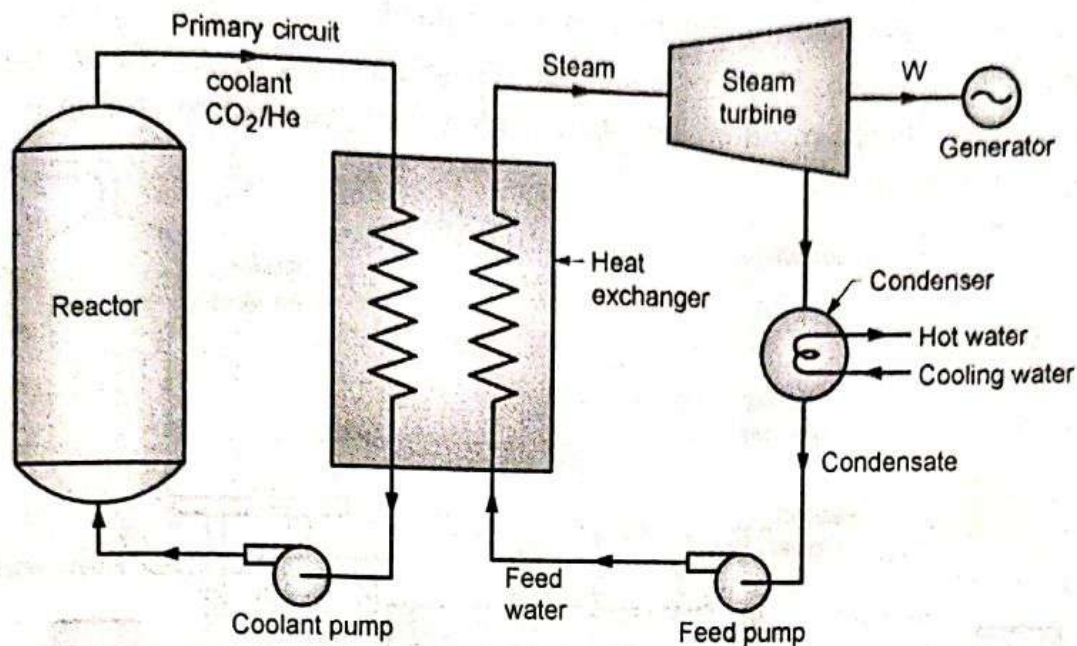


Fig. 10.17.1 : Gas cooler reactor power plant

As discussed above, coolant used is CO₂ for (GCGM) reactor and He in case of HTGC reactor in the primary circuit. The coolant transfers the heat energy to feed water in the heat exchanger. The steam so generated is used in conventional power plant to generate power working on Rankine cycle.

10.18 CANDU Reactor Power Plant : (Not in the syllabus: optional)

A reactor developed and designed by Canadian is called as CANDU (Canadian Deuterium Uranium) reactor.

It uses **pressurized heavy water (PHW)** (which is 99.8% deuterium oxide, D₂O) as moderator and primary coolant while the fuel used is natural uranium.

A CANDUPHW (Canadian Deuterium Uranium Pressurized Heavy Water) power plant is shown in Fig. 10.18.1.

Natural uranium oxide fuel is in the form of small cylinder pellets. These are packed in a corrosion resistant zirconium alloy tubes 0.5 cm long and 1.3 cm diameter to form a fuel rod. These short rods are combined in 37 bundles of 37 rods and 12 bundles are placed end to end in each pressure tubes. This type of arrangement helps in refuelling the reactor while in operation.

Reactor vessel is a steel cylinder called calandria. It is placed horizontally. It has pressure tubes penetrating the reactor vessel. The active core is about 6 m high and 7 to 8 m in diameter.

In the primary circuit, the D_2O coolant enters the array of pressure tubes at 110 bar and $260^\circ C$. It flows through the fuel element and leaves the pressure tubes at about $370^\circ C$ after absorbing the heat generated by fission of fuel material.

The coolant at 110 bar, $370^\circ C$ leaving the reactor enters the steam generator where the generated steam is used in conventional steam power plant.

Control rods are made up of *cadmium*. These are used to *start and shut down* the reactor. In addition there are other *absorbing rods* which are used to control power output during reactor operation.

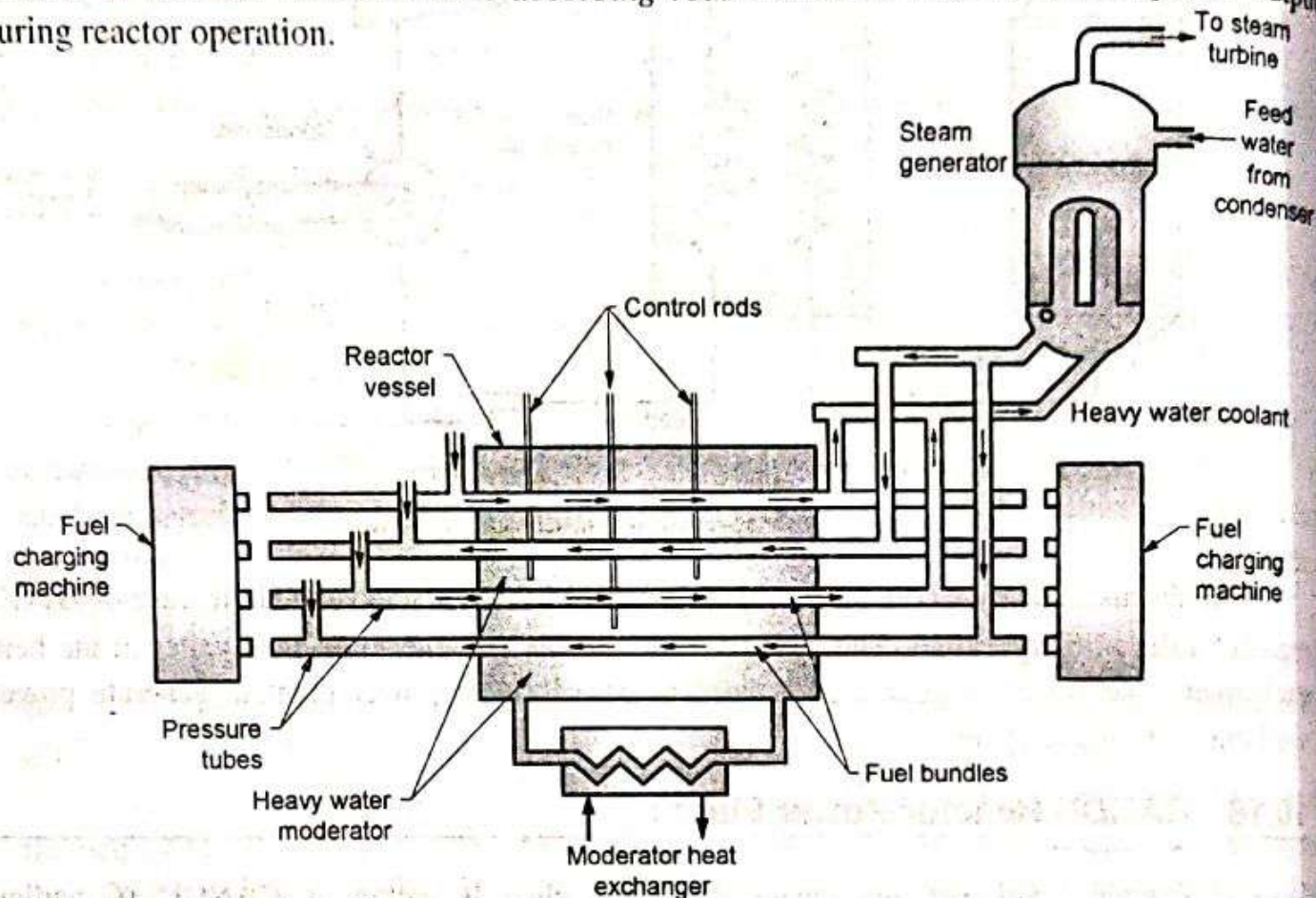


Fig. 10.18.1 : CANDU reactor

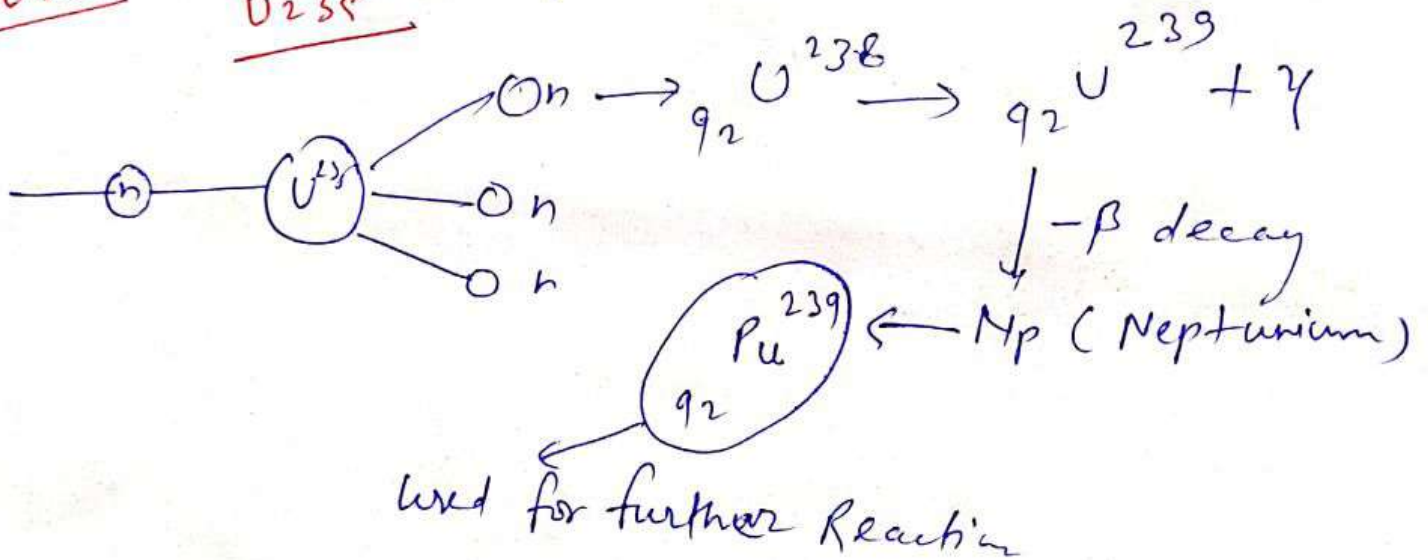
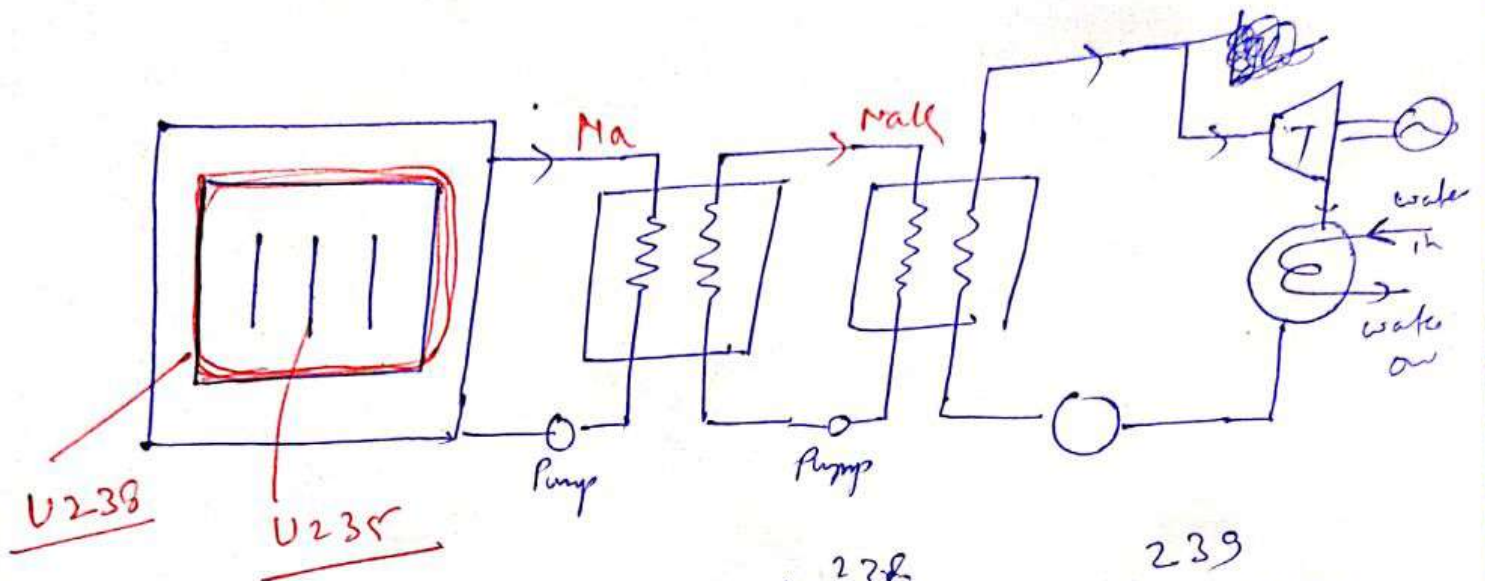
Advantages of CANDU reactor :

1. Heavy water is used as moderator which has low fuel consumption.
2. Enriched fuel is not required.
3. Cost and time of construction is less.
4. It has good neutron economy resulting into good breeding ratio.

Disadvantages of CANDU reactor :

1. Heavy water used is costly.
2. It has critical temperature limitations.
3. Leakage problems may occur.
4. Size of plant is large.
5. It requires high standards of design, manufacture and maintenance.

FAST Breeder Reactor!



10.19 Liquid Metal Reactor or Sodium Graphite Reactor (SGR) Power Plant :

Sodium graphite reactor is a typical liquid metal reactor. The arrangement of a sodium graphite reactor power plant is shown in Fig. 10.19.1.

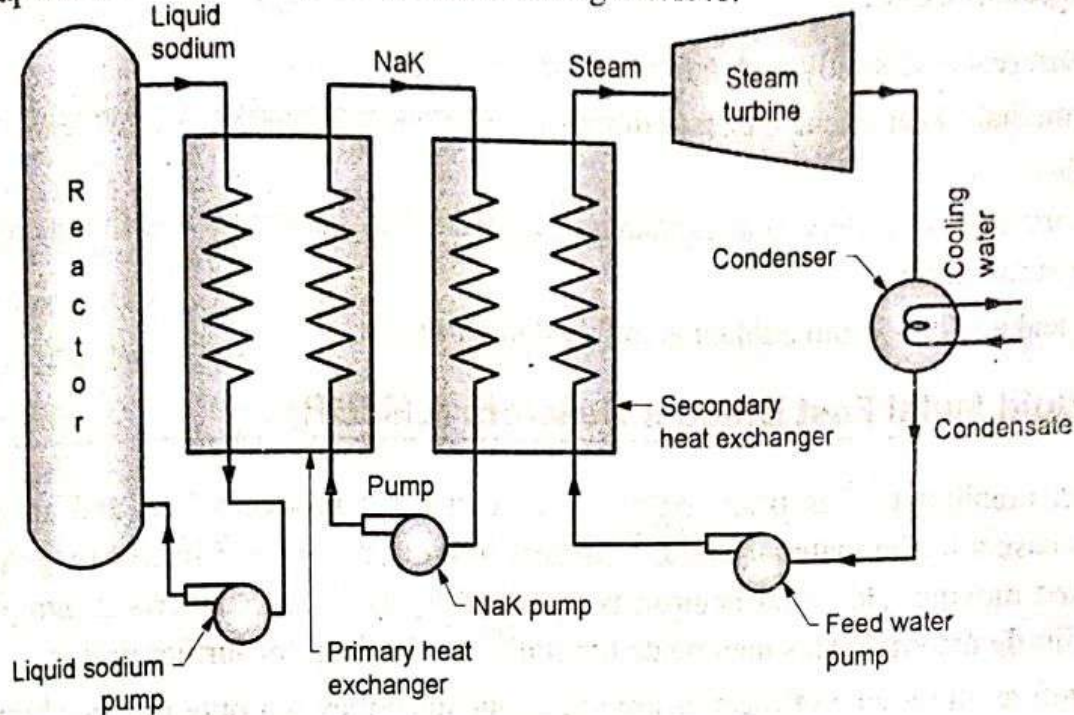


Fig. 10.19.1 : Sodium graphite reactor

It uses graphite as the moderator and liquid sodium as coolant which can reach a temperature of about 850°C at low pressure of only 7 bar.

In the primary circuit the heat is absorbed by liquid sodium in the reactor.

The sodium becomes radioactive while it passes through the core and reacts chemically with water.

Therefore, the heat absorbed by sodium is transferred to *secondary coolant sodium potassium (NaK)* in the *primary heat exchanger* which in turn transfers the heat in the secondary heat exchanger called **steam generator**.

Water leaving the steam generator is converted into superheated steam upto a temperature of 540°C . This steam is used for power generation in the steam plant circuit in the usual manner.

The reactor vessel, primary circuit and the primary heat exchanger have to be shielded from radiations.

The liquid metal is required to be handled under the cover of an inert gas like helium to prevent contact with air while charging or draining in the primary and secondary heat exchangers.

Advantages of SGR :

1. High temperatures of steam can be obtained due to use of liquid sodium as coolant.
2. System need not be pressurized.

10.21 Selection of Site for Nuclear Power Plants :

Various points to be considered in selection of site for nuclear power plants are as follows :

1. **Availability of water** for steam generation and cooling water for condenser. It should be nearer to sea, river or reservoir.
2. **Nearer to load centre** to reduce power transmission losses.
3. **Away from populated area** to safeguard the people from hazardous radioactive radiations.
4. **Availability of transport facility** for transportation of people and material at the time of its construction by road and rail.
5. **Safeguard against earthquakes** : The site should be away from seismic zone.
6. **Radioactive waste disposal** : The wastes of nuclear power plants being radioactive, the site should have sufficient space near the plant for its disposal.
7. **Soil conditions for foundation** : The bearing capacity of soil should be high so as to support heavy reactors on its foundations. The bearing capacity must be atleast 50 N/cm^2 .

**(Not in the syllabus:
Optional)**

Nuclear Waste Disposal : (Not in the syllabus: Optional)

Waste disposal of nuclear power station is of prime importance since the nuclear waste is likely to have radioactivity. Therefore, these wastes are disposed off in such a manner that it does not cause harm to human or plant life. The various methods adopted are :

- (i) In case of gaseous wastes, it is passed through filters and discharged at a high level through stacks.
- (ii) Moderate liquid wastes can be discharged after filtration, preliminary treatment (its PH value is adjusted) and by diluting and mixing with cooling water discharge into deep pits or dry wells.
- (iii) Highly radioactive liquid wastes are kept in concrete tanks and buried into ground till their decay of radioactivity.
- (iv) Solid wastes arising from discarded control rods, fuel cans etc. ~~are stored in shielded~~ concrete vaults.
- (v) The combustible and chemically incompatible wastes are regrettably. The combustible waste is burnt in incinerators and the flue gases formed are filtered and disposed off through stacks.

Active solid wastes are stored in water for about 100 or more days to allow radioactivity to decay. Then these are disposed off to deep salt mines or on ocean floor or in deep wells drilled in stable geological strata.

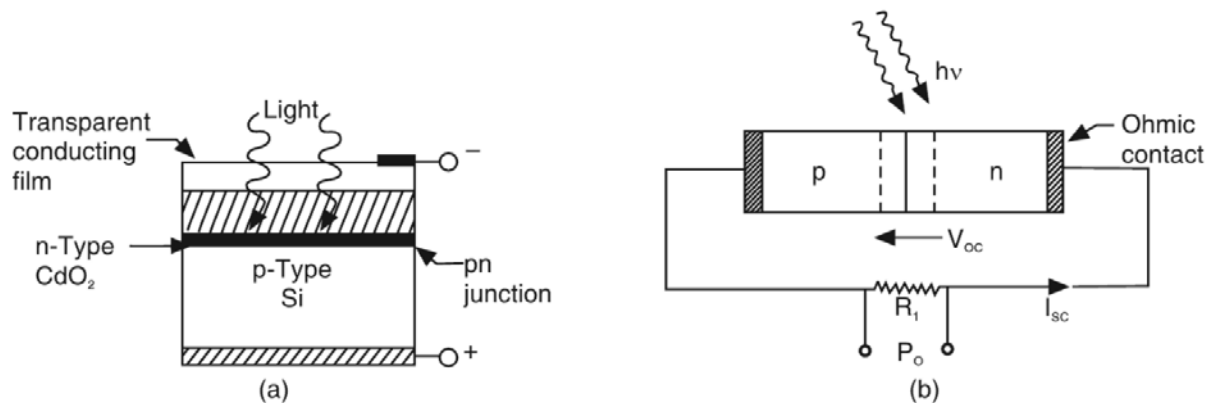
10.24 Comparison between Nuclear and Conventional Thermal Power Plants :

1. Advantages of nuclear power plants over thermal power plants :

- (i) For similar capacity plants, the space required is less.
- (ii) Cost of fuel transportation, storage and handling is very less since nuclear fuel requirements is much less compared to coal.
- (iii) It is more economical to operate, particularly in the areas remote to coal field.
- (iv) Ash handling problem is avoided.
- (v) Number of persons needed for operation are less.
- (vi) These are more reliable in operation.
- (vii) The capital cost is low for large sized power plants and the running cost is competitive.
- (viii) It has better performance at high load factors.

SOLAR CELL

A solar cell is basically a p-n junction that can generate electrical power, when illuminated. Solar cells are usually large area devices typically illuminated with sunlight and are intended to convert the solar energy into electrical energy.



The schematic of a solar cell is shown in above Fig. (a). It consists of a p-type chip on which a thin layer of n-type material is grown. When the solar radiation is incident on the cell, electron-hole pairs are generated in the n and p regions. The majority of them cannot recombine in the regions. They reach the depletion region at the junction where an electric field due to the space charge separates them. Electrons in the p-region are drawn into the n-region and holes in the n-region are drawn into the p-region. It results in accumulation of charge on the two sides of the junction and produces a potential difference called **photo emf**. Its magnitude is of the order of 0.5 V. The overall power-conversion efficiency of single-crystalline solar cells ranges from 10 to 30% yielding 10 to 30 mW/cm². If a load is connected across the cell a current flows through it. The sign convention of the current and voltage is shown in above Fig. (b). It considers a current coming out of the cell to be positive as it leads to electrical power generation. The power generated depends on the solar cell itself and the load connected to it.

1. Solar Cell I-V Characteristics:

The p - n Junction Diode: Understanding the I-V characteristics of a Solar cell is very important and for that it is desirable to quickly recapitulate the I-V characteristics of a simple P-N junction diode given in the figure below.

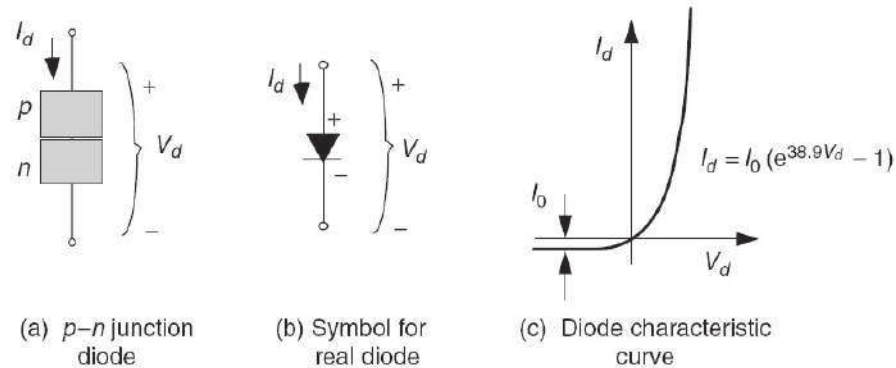


Figure-1 : A p - n junction diode allows current to flow easily from the p -side to the n -side, but not in reverse. (a) p - n junction (b) its symbol (c) its characteristic curve.

If we apply a voltage V_d across the diode terminals with polarity as shown, a forward current would flow through the diode from the p -side to the n -side. But if we try to send current in the reverse direction, only a very small ($\approx 10^{-12}$ A/cm²) reverse saturation current I_0 will flow. This reverse saturation current is the result of thermally generated carriers with the holes being swept into the p -side and the electrons into the n -side. In the forward direction, the voltage drop across the diode is only a few tenths of a volt.

The voltage - current characteristic curve for the p - n junction diode is given by the following Shockley diode equation:

$$I_d = I_0 (e^{qV_d/kT} - 1)$$

where I_d is the diode current in the direction of the arrow (A), V_d is the voltage across the diode terminals from the p -side to the n -side (V), I_0 is the reverse

saturation current (A), q is the electron charge (1.602×10^{-19} C), k is Boltzmann's constant (1.381×10^{-23} J/K), and T is the junction temperature (K).

Substituting the above constants into the exponent of equation above for I_d gives

$$\frac{qV_d}{kT} = \frac{1.602 \times 10^{-19}}{1.381 \times 10^{-23}} \cdot \frac{V_d}{T(\text{K})} = 11,600 \frac{V_d}{T(\text{K})}$$

A junction temperature of 25°C is often used as a standard, which when substituted in the Shockley diode equation gives the following diode current I_d .

$$I_d = I_0(e^{38.9V_d} - 1) \quad (\text{at } 25^\circ\text{C})$$

Example-1: A $p-n$ Junction Diode. Consider a $p-n$ junction diode at 25°C with a reverse saturation current of 10^{-9} A. Find the voltage drop across the diode when it is carrying the following currents:

- a) No current (open-circuit voltage) b) 1 A c) 10 A

Solution:

a) In the open-circuit condition, $I_d = 0$, so from equation for I_d we get $V_d = 0$.

b) With $I_d = 1$ A, we can find V_d by rearranging the above equation for ' I_d ' :

$$e^{38.9V_d} = \left(\frac{I_d}{I_0} + 1 \right)$$

Taking natural logarithms (to the base 'e') on both side and then dividing both sides by 38.9 we get

$$V_d = \frac{1}{38.9} \ln \left(\frac{I_d}{I_0} + 1 \right) = \frac{1}{38.9} \ln \left(\frac{1}{10^{-9}} + 1 \right) = 0.532 \text{ V}$$

c) with $I_d = 10$ A,

$$V_d = \frac{1}{38.9} \ln \left(\frac{10}{10^{-9}} + 1 \right) = 0.592 \text{ V}$$

As can be seen, the voltage drop changes very little as the diode conducts more and more current, changing by only about 0.06 V as the current increased by a factor of 10. Often in normal electronic circuit analysis, the diode voltage drop when it is conducting rated current is about 0.6 V, which is quite in line with the above results.

A Simple Equivalent Circuit of a Photovoltaic Cell:

To derive the I-V characteristics of a Solar cell a simple equivalent circuit depicting all its characteristics is developed. It consists of a real PN Junction diode in parallel with an ideal current source as shown in the figure below. The ideal current source delivers current proportional to the solar '*Insolation*' to which it is exposed.

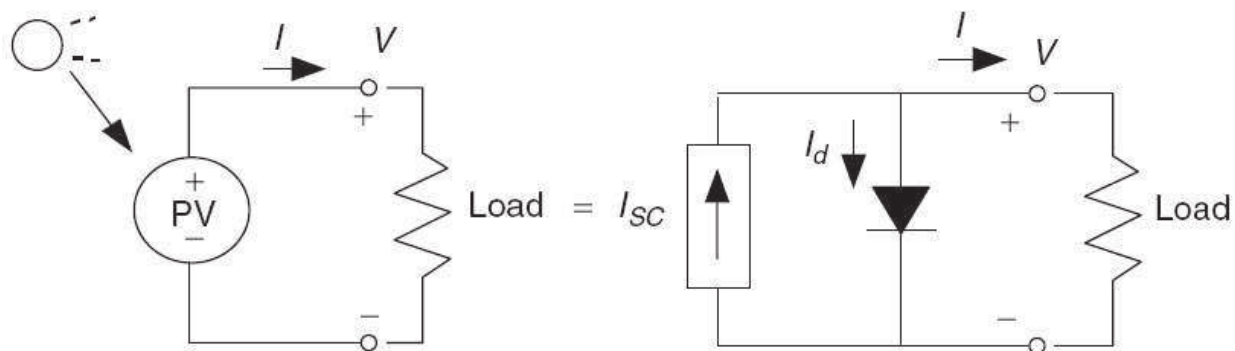


Figure-2: A simple equivalent circuit of a photovoltaic (Solar) cell consists of a current source driven by sunlight in parallel with a real diode.

There are two important conditions for the actual Solar cell in its equivalent circuit as shown in the figure below (Figure -3). They are:

(a) The current that flows when the terminals are shorted together (the short-circuit current, I_{sc}). When the leads of the equivalent circuit for the PV cell are shorted together, no current flows in the (real) diode since $V_d = 0$. That means $I_d =$

0 and all the current from the ideal source I_{sc} flows through the shorted leads. Since that short-circuit current is I_{sc} , the magnitude (maximum value) of the ideal current source itself is equal to I_{sc} .

(b) The voltage across the terminals when the leads are left open (the open-circuit voltage, V_{oc}). Since output terminals are open no current flows in the load i.e. $I = 0$

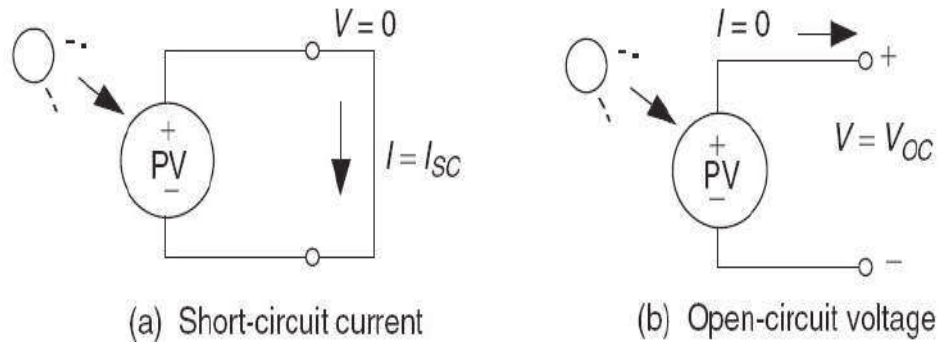


Figure-3: Two important parameters for a Solar cell are the short-circuit current I_{sc} and the open-circuit voltage V_{oc} .

Now we can write a voltage and current equation for the equivalent circuit of the PV cell shown in figure (b) starting with

$$I = I_{sc} - I_d$$

and then substituting the value of I_d as per the Shockley equation into the above equation for I we get :

$$I = I_{sc} - I_0 (e^{qV/kT} - 1)$$

It is interesting to note that the second term in the above equation is just the diode equation with a negative sign. That means that a plot of the above equation is just I_{sc} added to the diode characteristic (in the first quadrant) turned upside-down. Figure below shows the current–voltage relationship for a PV cell when it is dark (no illumination) and light (with illumination) based on the above equation.

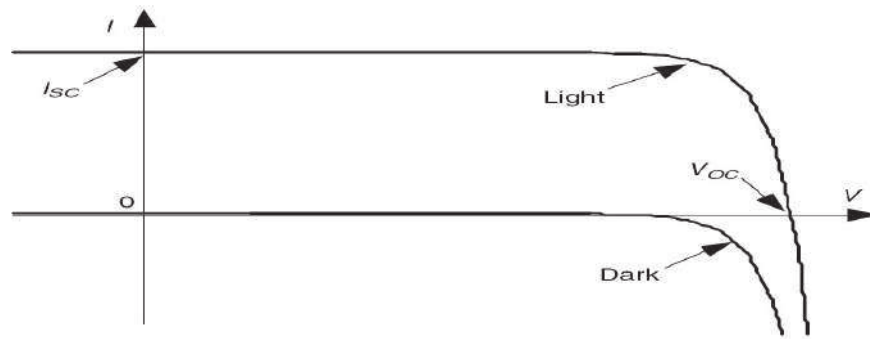


Figure-4: Photovoltaic current–voltage relationship for “dark” (no sunlight) and “light” (an illuminated cell). The dark curve is just the diode curve turned upside-down. The light curve is the dark curve plus I_{sc} .

When the leads from the PV cell are left open, $I = 0$ and we can solve the above equation (8.8) for the open-circuit voltage V_{oc} :

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{I_{sc}}{I_0} + 1 \right)$$

And at 25°C , both the above equations (8.8) and (8.9) become

$$I = I_{sc} - I_0(e^{38.9 V} - 1)$$

And

$$V_{oc} = 0.0257 \ln \left(\frac{I_{sc}}{I_0} + 1 \right)$$

In both of the above equations (9&10) , short-circuit current, I_{sc} , is directly proportional to solar ‘**Insolation**’, which means that we can now quite easily plot sets of PV current–voltage curves for varying sunlight for any given Solar Cell. Also, quite often laboratory specifications for the performance of photovoltaics are given per cm^2 of junction area, in which case the currents in the above equations are written as current densities. Both these points are illustrated in the following example.

Example-2: I –V Curve of a Photovoltaic Cell. Consider a 100-cm² photovoltaic cell with reverse saturation current $I_0 = 10^{-12}$ A/cm². In full sun, it produces a short-circuit current of 40 mA/cm² at 25°C. Find the open-circuit voltage at full sun and again for 50% sunlight. Plot the results.

Solution: The reverse saturation current of the cell $I_0 = 10^{-12}$ A/cm² × 100 cm² = 1 × 10⁻¹⁰ A. At full sun I_{SC} is 0.040 A/cm² × 100 cm² = 4.0 A. From (8.11) the open-circuit voltage @ full sun light is:

$$V_{OC} = 0.0257 \ln \left(\frac{I_{SC}}{I_0} + 1 \right) = 0.0257 \ln \left(\frac{4.0}{10^{-10}} + 1 \right) = 0.627 \text{ V}$$

Since short-circuit current is directly proportional to solar intensity, at half sun $I_{SC} = 2$ A and the open-circuit voltage @ half sun light is:

$$V_{OC} = 0.0257 \ln \left(\frac{2}{10^{-10}} + 1 \right) = 0.610 \text{ V}$$

Plot of the current using the equation $I = I_{SC} - I_0(e^{38.9 V} - 1)$ @ full sun light @ half sun light along with the corresponding open circuit voltages is shown below.

It can be noticed that with reduced Sunlight intensity only I_{SC} gets reduced correspondingly but the reduction in V_{OC} is quite small (0.627 to 0.61 V)

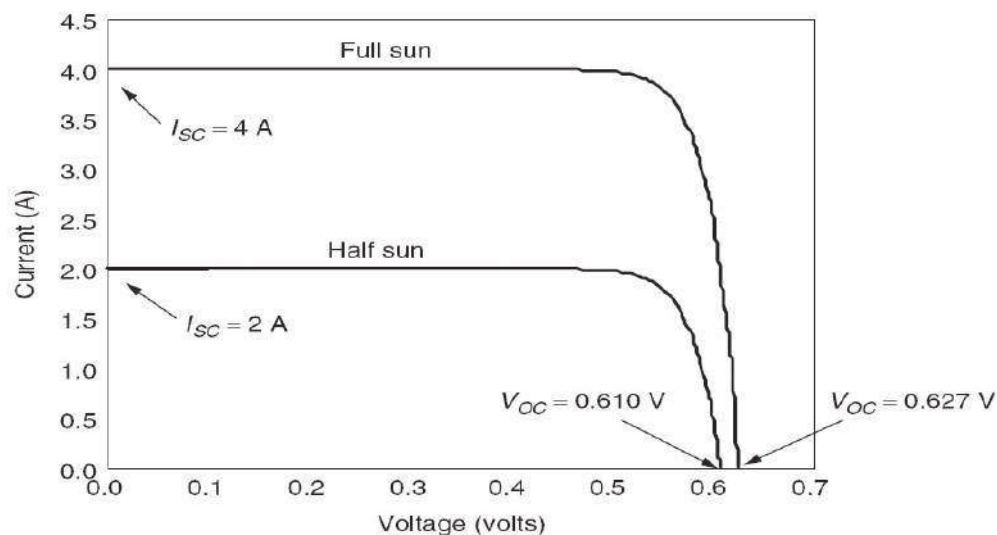


Figure-5: I-V Plot corresponding to the above Problem

2. Maximum Power Point

Let us consider, for the moment, a single PV module connected to a load as shown in the figure below. The load might be a dc motor driving a pump or it might be a battery, for example. Before the load is connected, the module sitting in the sun will produce an open-circuit voltage V_{OC} , but no current will flow. If the terminals of the module are shorted together (which doesn't damage the module at all, by the way), the short-circuit current I_{SC} will flow, but the output voltage will be zero. In both cases, since power is the product of current and voltage, no power is delivered by the module and no power is received by the load. When a load is connected, some combination of current and voltage will result and power will be delivered. To figure out how much power is delivered, we have to consider the **I-V** characteristic curve of the **module** as well as the **I-V** characteristic curve of the **load**.

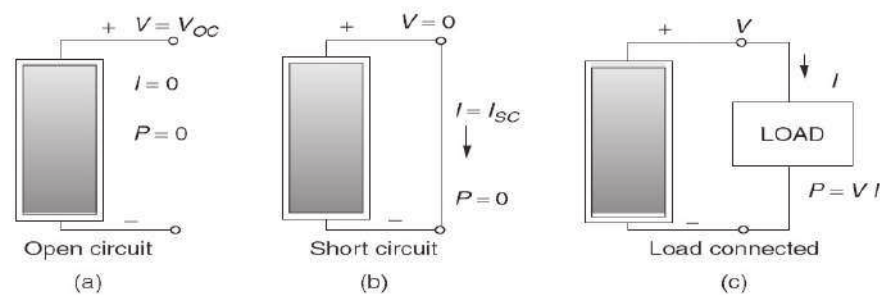


Figure-6: No power is delivered when the circuit is open (a) or shorted (b). When the load is connected (c), the same current flows through the load and module and the same voltage appears across them.

Figure-7 below shows a generic **I-V** curve for a PV module, identifying key parameters like open-circuit voltage V_{OC} and short-circuit current I_{SC} that we have explained. Also shown is the product of voltage and current, i.e., the power delivered by the module. At the two ends of the **I-V** curve, the output power is zero since either current or voltage is zero at those points. The '**Maximum Power Point**' (MPP) is that spot near the knee of the **I-V** curve at which the product of current and voltage reaches its maximum. The voltage and current at the MPP are designated as V_m and I_m in general and as V_R and I_R (*rated voltage* and *rated current*) under idealized test conditions.

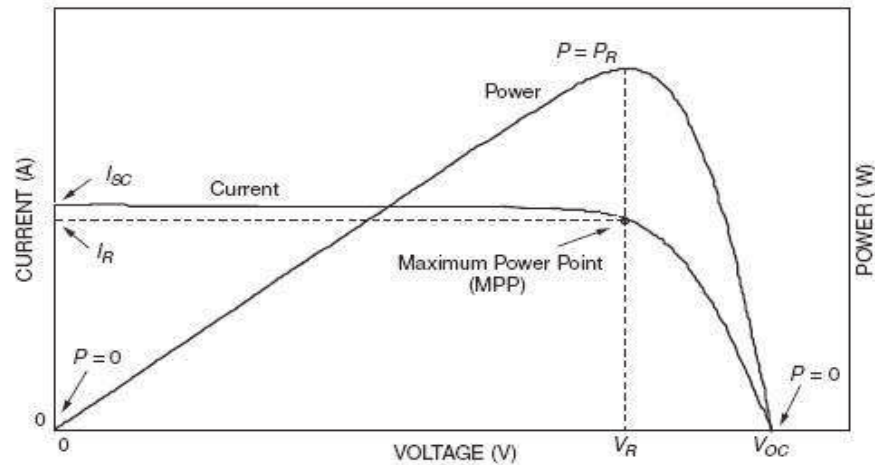


Figure-7: The I –V curve and power output for a PV module. At the maximum power point (MPP) the module delivers the Maximum Power that it can under the given conditions of sunlight and temperature .

Another way to visualize the location of the *Maximum Power Point* is by trying to find the biggest possible rectangle that will fit beneath the **I-V** curve. As shown in the figure-8 below the sides of the rectangle correspond to current and voltage, and so its area is power.

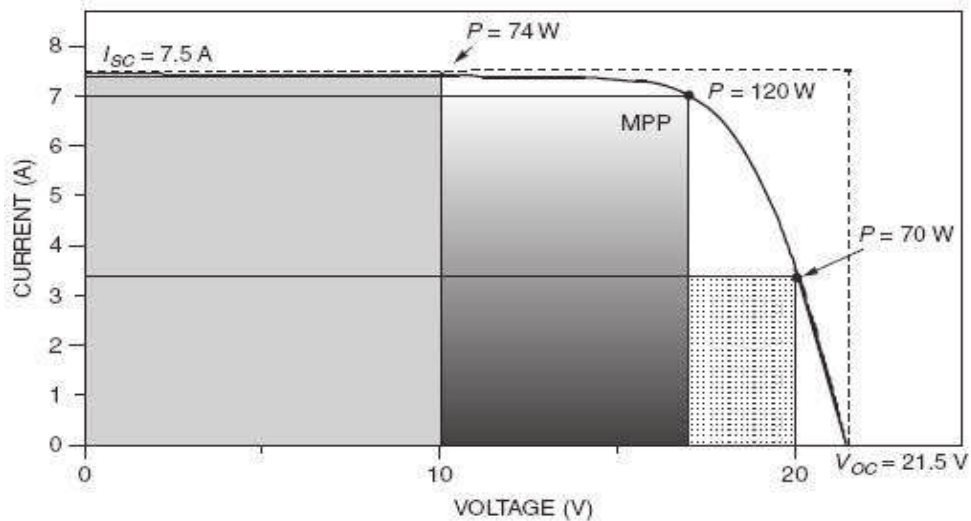


Figure-8: The Maximum Power Point (MPP) corresponds to the biggest rectangle that can fit beneath the I –V curve. The fill factor (FF) is the ratio of the area (power) at MPP to the area formed by a rectangle with sides V_{OC} and I_{SC} .

The italicized material given below between the marked lines is not in syllabus. Furnished just for information only.

Since PV I –V curves shift all around as the amount of insolation changes and as the temperature of the cells varies, standard test conditions (STC) have been established to enable fair comparisons of one module to another. Those test conditions include a solar irradiance of 1 kW/m^2 (1 sun) with spectral distribution Shown in 'unit-1 Fig-22', corresponding to an air mass ratio of 1.5 (AM 1.5). The standard cell temperature for testing purposes is 25°C (it is important to note that 25° is cell temperature, not ambient temperature). Manufacturers always provide performance data under these operating conditions, some examples of which are shown in table below. The key parameter for a module is its rated power; to help us remember that it is dc power measured under standard test conditions, it has been identified in the table below as PDC, STC. Later we'll learn how to adjust rated power to account for temperature effects as well as see how to adjust it to give us an estimate of the actual ac power that the module and inverter combination will deliver.

Table: Examples of PV Module Performance Data Under Standard Test Conditions (1 kW/m^2 , AM 1.5, 25°C Cell Temperature)

Manufacturer	Kyocera	Sharp	BP	Uni-Solar	Shell
Model	KC-120-1	NE-Q5E2U	2150S	US-64	ST40
Material	Multicrystal	Polycrystal	Monocrystal	Triple junction a-Si	CIS-thin film
Number of cells n	36	72	72		42
Rated Power $P_{\text{DC,STC}}$ (W)	120	165	150	64	40
Voltage at max power (V)	16.9	34.6	34	16.5	16.6
Current at rated power (A)	7.1	4.77	4.45	3.88	2.41
Open-circuit voltage V_{OC} (V)	21.5	43.1	42.8	23.8	23.3
Short-circuit current I_{SC} (A)	7.45	5.46	4.75	4.80	2.68
Length (mm/in.)	1425/56.1	1575/62.05	1587/62.5	1366/53.78	1293/50.9
Width (mm/in.)	652/25.7	826/32.44	790/31.1	741/29.18	329/12.9
Depth (mm/in.)	52/2.0	46/1.81	50/1.97	31.8/1.25	54/2.1
Weight (kg/lb)	11.9/26.3	17/37.5	15.4/34	9.2/20.2	14.8/32.6
Module efficiency	12.9%	12.7%	12.0%	6.3%	9.4%

3. Cell efficiency & Fill Factor

Cell Efficiency: It is defined as the ratio of maximum electrical power output to the radiation power input to the cell and it is expressed in percentage. It is considered that the radiation power on the earth is about 1000 watt/square meter hence if the exposed surface area of the cell is A then total radiation power on the cell will be 1000 A watts. Hence the efficiency of a solar cell may be expressed as:

$$Efficiency(\eta) = \frac{P_m}{P_{in}} \approx \frac{P_m}{1000A}$$

Fill factor: This is another quantity that is often used to characterize module performance. The *fill factor* (FF) is the ratio of the power at the maximum power point to the product of V_{oc} and I_{sc}

$$\text{Fill factor (FF)} = \frac{\text{Power at the maximum power point}}{V_{oc} I_{sc}} = \frac{V_R I_R}{V_{oc} I_{sc}}$$

So FF can be visualized as the ratio of two rectangular areas i.e. ratio of the rectangle area corresponding to MPP and the rectangle area corresponding to V_{oc} & I_{sc} (shown in dotted lines) as shown in the earlier figure .

Fill factors of around 70–75% for crystalline silicon solar modules are typical.

4. Effect of irradiation and temperature:

I-V curves shift with change in cell temperatures and Insolation. Manufacturers often provide $I - V$ curves that show how the curves shift as insolation and cell temperature changes. Figure below shows examples for a typical (**the Kyocera 120-W**) multi crystal (poly crystal) - silicon module described in the table above.

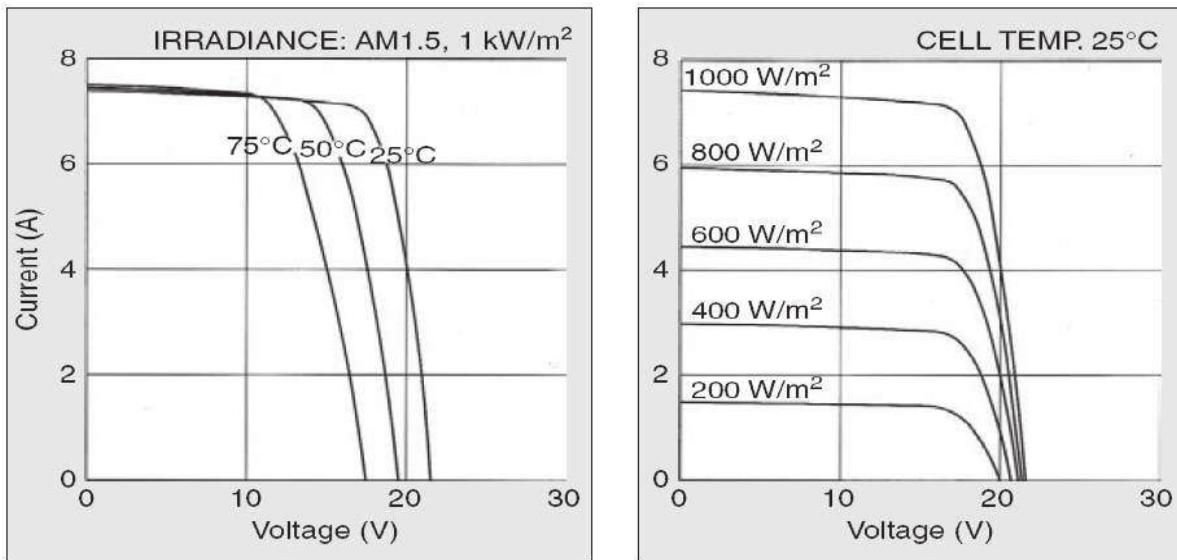


Figure-9: Current-voltage characteristic curves under various cell temperature and irradiance levels for a typical PV module.

As can be seen in the figure above, as cell temperature **increases**, the open-circuit voltage V_{oc} **decreases** substantially while the short-circuit current I_{sc} **increases** only slightly thus effectively decreasing the maximum power and efficiency. Solar cells therefore perform better on cold, clear days than hot ones. Due to this significant shift in performance as cell temperature changes, the temperature needs to be included in any estimate of module performance.

Cell temperature varies not only because of ambient temperature change, but also because of change in insolation on the cell. Since only a small fraction of the insolation hitting a module is converted to electricity and carried away, most of that excess incident energy is absorbed and converted into heat.

5. Principles of Maximum Power Point Trackers:

Maximum Power Point Tracking, frequently referred to as MPPT, operates Solar PV modules in a manner that allows the modules to produce all the power they are capable of generating. MPPT is not a mechanical tracking system but it works on a particular tracking algorithm and it is based on electronic control. However

MPPT can be used in conjunction with a mechanical tracking system, but the two systems are completely different. The voltage at which PV module produces maximum power is called 'Maximum Power Point' and this lies somewhere close to the knee point. We have already seen that this Maximum Power Point in the I-V characteristics of a Solar cell shifts with solar radiation and solar cell temperature (operating conditions).

We also know that the operating point of any power system i.e. the point at which maximum power is delivered by a source to a given load is the intersection point of the source line (in this case the I-V characteristic of the Solar Module) and the load line(V-I characteristic of the Load). Let us consider a PV source having the I-V and P-V characteristics as shown in figure-9 (a) & (b) below and supplying power to three different loads R_1, R_2 , and R_3 . As the load resistance increases from R_1 to R_2 the operating point shifts from A_1 to A_2 and when load resistance increases from R_2 to R_3 , the operating point moves from A_2 to A_3 . As can be seen the maximum power is delivered by the module to the load when the load resistance is R_2 . In figure (a) it is @ the knee point of the I-V curve and in figure (b) it is @ the peak power point of the P-V curve.

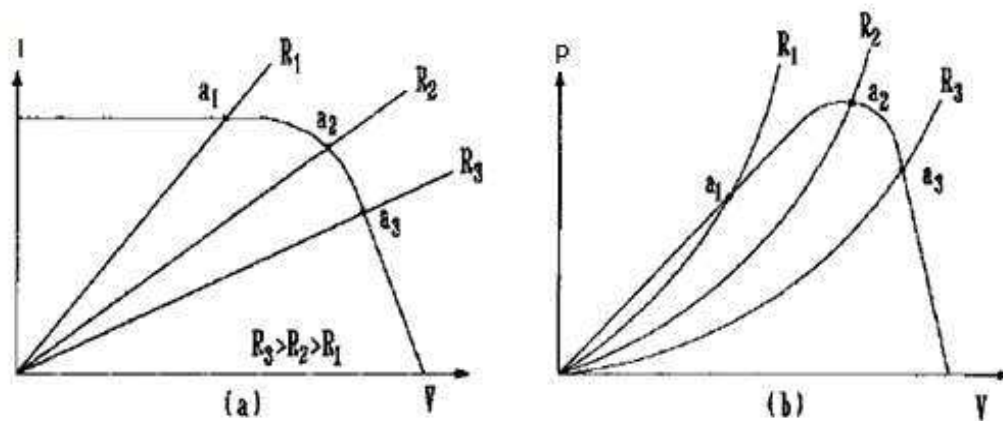


Figure-9: Demonstration of the point that Maximum Power Transfer From a given Solar cell @ given conditions takes place only with a specific load.

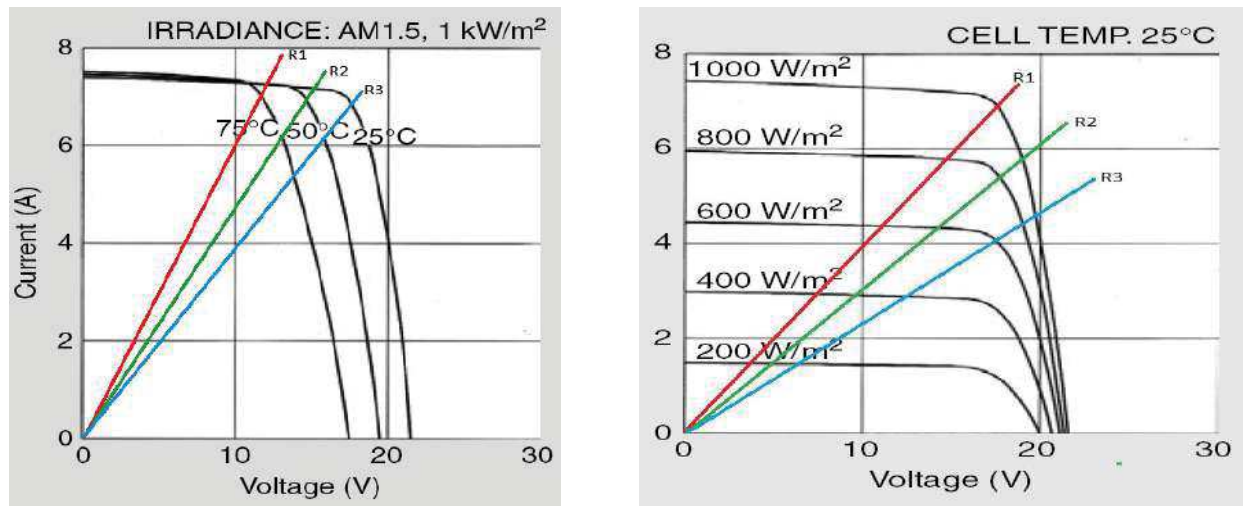


Figure-10: (a) Effect of Temperature and (b) Insolation along with load on the shift of the Maximum Power Point.

Further, when the temperature and solar radiation change along with load how the maximum power point shifts is shown in the figure-10 (a) and (b) above separately. These figures have been obtained by super imposing the different load lines on the earlier curves showing how the MPP varies with insolation and cell temperature. These figures clearly demonstrate how tricky it is to match the source with load with so many variables i.e. Cell temperature, Insolation and Load.

From the above characteristics it can be observed that since a PV cell has an exponential relationship between current and voltage at higher voltages, maximum power generation and maximum power transfer to the load occur at the knee of the curve, when simultaneously the load resistance (V/I) is equal to the negative of the differential resistance at that point of the solar cell I-V characteristic.

Such critical matching of Source with the Load in so many variable conditions is precisely what is done by the MPPT acting as an interface between the Source and the Load.

Thus, in summary a MPPT can be considered as a high-efficiency DC-to-DC converter that functions as an optimal electrical interface which:

-
1. Has to first search and then establish the Input DC Voltage from a Solar module corresponding to the Maximum power point based on the prevailing operating conditions. And then
 2. It has to convert this voltage into a DC Voltage again suitable to the varying Load conditions so as to extract the full power available from the Solar cell and deliver it to the load.

In order to carry out such complex functions Maximum power point trackers normally utilize microprocessor based controllers with sophisticated control algorithms.

Solar Power plants operate in both Off - Grid and ON - Grid modes. MPPTs make the best use of all the energy generated by the panels in both the cases as below:

- In the case of ON-Grid PV plants, MPPT is used to extract the maximum power from a PV array, convert this to alternating current (AC) and forward the excess energy to the power grid after catering to the local Load. If the power is less, then it draws the additional power from the Grid.
- In the case of OFF - Grid PV plants where battery storage is necessary, MPPTs are used as charge controllers to extract the full power from the Solar cell and distribute between the load and for charging the battery as necessary.

6. PV Modules and Arrays:

PV Modules: An individual cell produces about one watt of power at about 0.5 V and they are of no practical use. Typically, it is a few square inches in size. Hence the basic building block for PV applications is a **module** consisting of a number of pre-wired cells in series, all encased in tough, weather-resistant packages in an area of several square feet. Such an encased panel is called a **Solar Module**.

Most solar PV panels have 30 to 36 cells connected in series. Each cell produces about 0.5 V in sunlight, so a panel produces 15V to 18 V. These panels are designed to charge 12-V batteries. A 30-cell panel (15 V) can be used to charge the battery without a controller, but it may fail to charge the battery completely.

A 36-cell panel (18 V) will do better, but needs a controller to prevent overcharging.

The current depends on the size of each cell, and the solar radiation intensity. Most cells produce a current of 2 A to 3 A in bright sunlight. The current is the same in every cell because the cells are connected in series.

For cells wired in series, their voltages at any given current add. A typical module will have 36 cells.

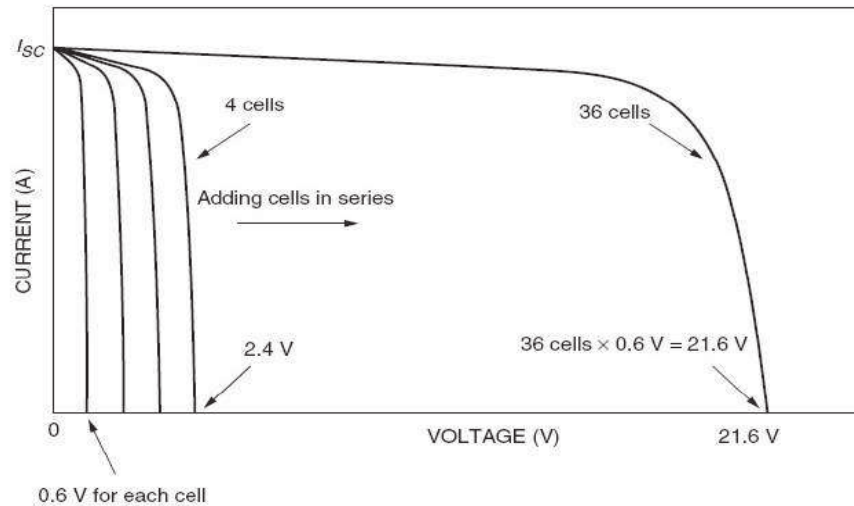


Figure-11: For cells wired in series, their voltages at any given current add. A typical module will have 36 cells.

Arrays: Multiple modules, in turn, can be wired in series to increase voltage and in parallel to increase current, the product of which is power.

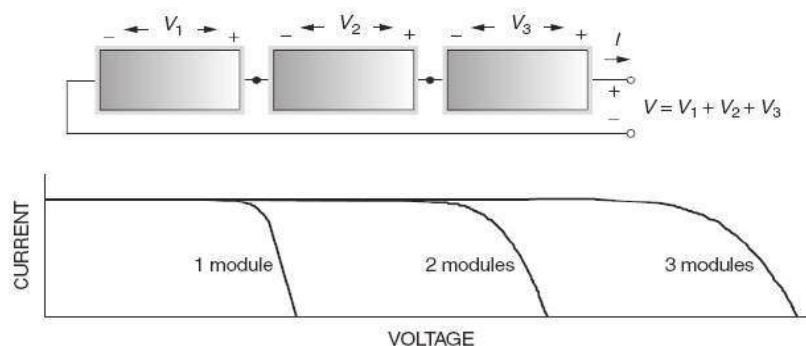


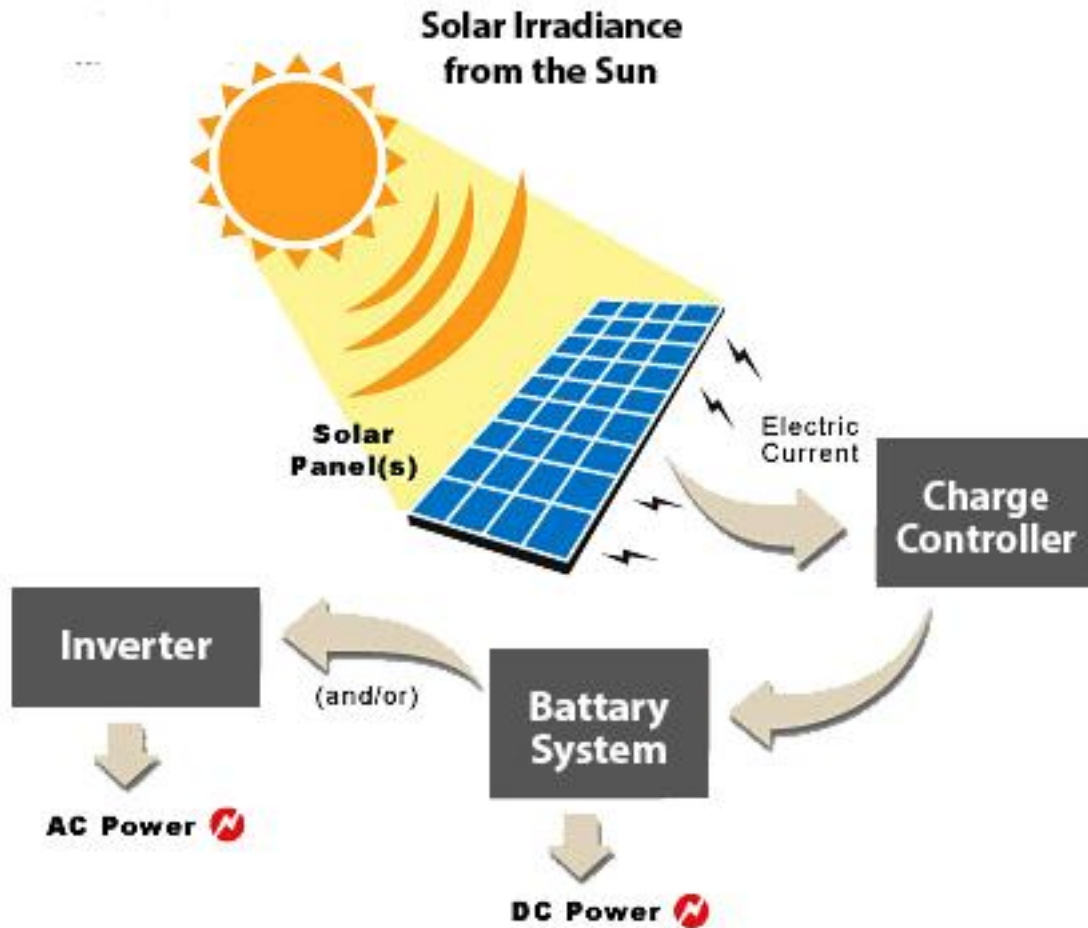
Figure-12: Modules in series, at any given current the voltages add

Introduction to renewable energy sources

Renewable energy sources derive their energy from existing flows of energy from on-going natural processes, such as sunshine, wind, flowing water, biological processes, and geothermal heat flows. A general definition of renewable energy sources is that renewable energy is captured from an energy resource that is replaced rapidly by a natural process such as power generated from the sun or from the wind. Currently, the most promising (aka economically most feasible) alternative energy sources include wind power, solar power, and hydroelectric power. Other renewable sources include geothermal and ocean energies, as well as biomass and ethanol as renewable fuels.

Solar

The recent disasters in the southeastern United States highlighted the decline in the world's oil supply, forcing us to begin considering other energy options. One promising technology, solar power is worth considering for its sustainable, renewable and emissions reducing qualities. Modern residential solar power systems use photovoltaic (PV) to collect the sun's energy. "Photo" means "produced by light," and "voltaic" is "electricity produced by a chemical reaction." PV cells use solar energy to generate a chemical reaction that produces electricity. Each cell contains a semiconductor; most commonly silicon in one of several forms (single-crystalline, multi-crystalline, or thin-layer), with impurities (either boron or phosphorus) diffused throughout, and is covered with a silk screen. Cells are joined together by a circuit and frame into a module. Semiconductors allow the electrons freed from impurities by the sun's rays to move rapidly and into the circuit, generating electricity. Commercial residential PV modules range in power output from 10 watts to 300 watts, in a direct current. A PV module must have an inverter to change the DC electricity into alternating current energy in order to be usable by electrical devices and compatible with the electric grid. PV modules can also be used en masse to create large-scale power plants.

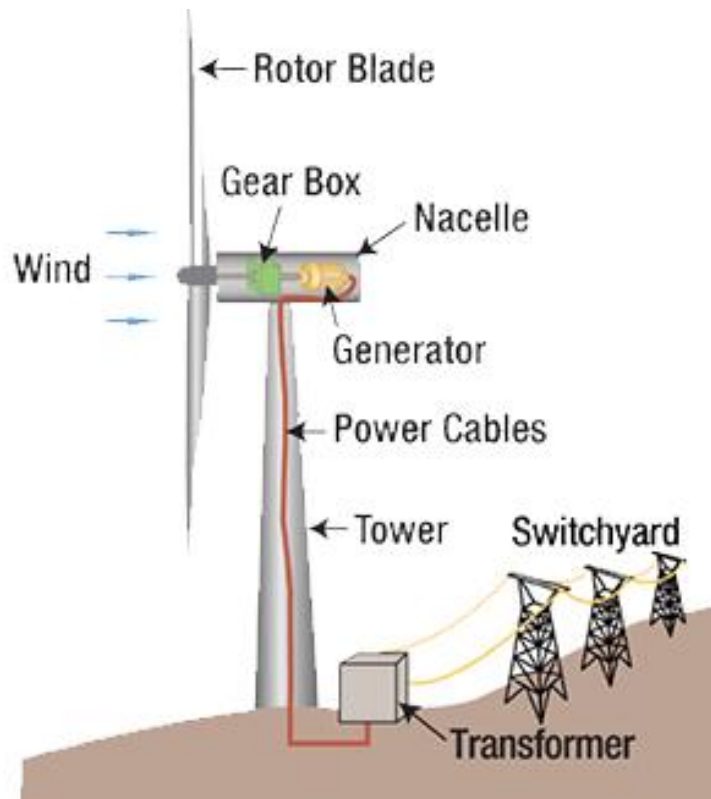


Using PV modules to generate electricity can significantly reduce pollution. The most energy used in creating solar panels is used to purify and crystallize the semiconductor material. No official numbers are available on the exact amount of energy used to create solar panels because there is no industry standard for making the crystals. A number of researchers have done work in attempt to address concerns about energy payback for PV systems. Assuming 12% conversion efficiency and 1,700 kWh/m² of sunlight per year, the estimates range between 2 and 4 years for rooftop PV systems to generate the energy it took to make them.⁴⁴ The average United States household uses 830 kWh of electricity per month. Over twenty years, a 100-megawatt solar thermal electric power plant can avoid producing over three million tons of carbon dioxide. Estimates regarding pollution prevention suggest that producing 1,000 kWh of electricity through solar power can reduce emissions by 8 pounds of sulfur dioxide, 5 pounds of nitrogen oxide, and 1,400 pounds of carbon dioxide. Lifetime estimates (over a projected 28 years) average in the thousands of pounds of prevented emissions. Installing a PV system is a hefty

investment for homeowners. 5-kW systems can cost up to \$40,000. PV system power can cost as much as \$9 per watt, and small systems will not produce enough power to offset electricity costs and save the homeowner any substantial money. As a result, over 30 states offer incentives (mostly in the form of tax rebates) to help encourage homeowners to purchase and install PV systems. California is one of the key states, receiving a huge amount of radiation, with the better part of the industry located there, and high-energy costs. The state of California offers a number of incentives under the Emerging Renewables Program passed by the state legislature. The California Energy Commission offers a rebate of \$4 per watt to help homeowners affordably install PV systems. However, the mortgage financing required to purchase a realistic PV system is still quite substantial. As PV technology advances, more efficient, easily affordable, standardized, reliable and longer-lasting modules will become available. PV systems' value to the energy sector especially in residential capacities, is increasingly apparent. However, the continued high cost means that many homeowners will be deterred from purchasing and installing PV systems. The only way to encourage further growth in this sector is for consumers to purchase such systems. The energy emissions reductions are substantial enough to be worth the consideration of the federal government. In order to encourage consumers' interest in PV systems and growth in the renewable energy sector at a faster rate, the federal government should create an incentive program to help homeowners and businesses purchase and install PV systems, especially on new constructions.

Wind

Wind energy is one of the most promising alternative energy technologies of the future. Throughout recent years, the amount of energy produced by wind-driven turbines has increased exponentially due to significant breakthroughs in turbine technologies, making wind power economically compatible with conventional sources of energy. Wind energy is a clean and renewable source of power. The use of windmills to generate energy has been utilized as early as 5000 B.C., but the development of wind energy to produce electricity was sparked by the industrialization. The new windmills, also known as wind turbines, appeared in Denmark as early as 1890. The popularity of wind energy however has always depended on the price of fossil



fuels. For example, after World War II, when oil prices were low, there was hardly any interest in wind power.

However, when the oil prices increased dramatically in the 1970s, so did worldwide interest in the development of commercial use of electrical wind turbines. Today, the wind-generated electricity is very close in cost to the power from conventional utility generation in some locations.

Where does wind come from? Wind

is a form of solar energy and is caused by the uneven heating of the atmosphere by the Sun, the irregularities of the Earth's surface, and rotation of the Earth. The amount and speed of wind depends on the Earth's terrain and other factors. The wind turbines use the kinetic energy of the wind and convert that energy into mechanical energy, which in turn can be converted into electricity by means of a generator. 2

There are essentially two types of wind turbines: The horizontal-axis variety, and the vertical-axis design. The horizontal-axis design is used more commonly and looks like an Old Dutch windmill, whereas the vertical-axis design looks like an eggbeater. These wind turbines generally have either two or three blades, called rotors, which are angled at a pitch to maximize the rotation of the rotors. The horizontal-axis design is slightly more efficient and dependable than the vertical-axis windmill. Most of the windmill models that are currently in production are thus horizontal-axis windmills.

Utility scale turbines can produce anywhere from 50 kilowatts to several megawatts of energy. These large windmills are generally grouped together in a windy area in what is called a wind

farm. The proximity of the windmills in a wind farm makes it easier to feed the produced electricity into the power grid. Wind energy offers many advantages compared to fossil based power and even some other types of alternative energy, which explains why it is the fastest growing energy source in the world. The two main reasons are cleanliness and abundance. The fact that wind is a renewable resource gives it a major advantage over oil and the nonrenewable resources. Considering that environmental pollution is being linked to several global problems that might eventually threaten the existence or at the very least worsen human living conditions, the fact that windmills do not produce any emissions whatsoever is another reason to increase the use of wind turbines. Increasing the percentage of wind power used by the United States would not be unreasonable, seeing that the price of wind power is between 4 and 6 cents.

Even though wind energy has many environmental and supply advantages, there are several disadvantages that limit the usability of wind power. The main disadvantage to wind power is that it is unreliable. Wind does not blow at a constant rate, and it does not always blow when energy is needed. Furthermore, the windiest locations are often in remote locations, far away from big cities where the electricity is needed. Just like with any other energy plant, people oppose it because of aesthetic reasons. The rotor noise produced by the rotor blades is another reason for opposition.

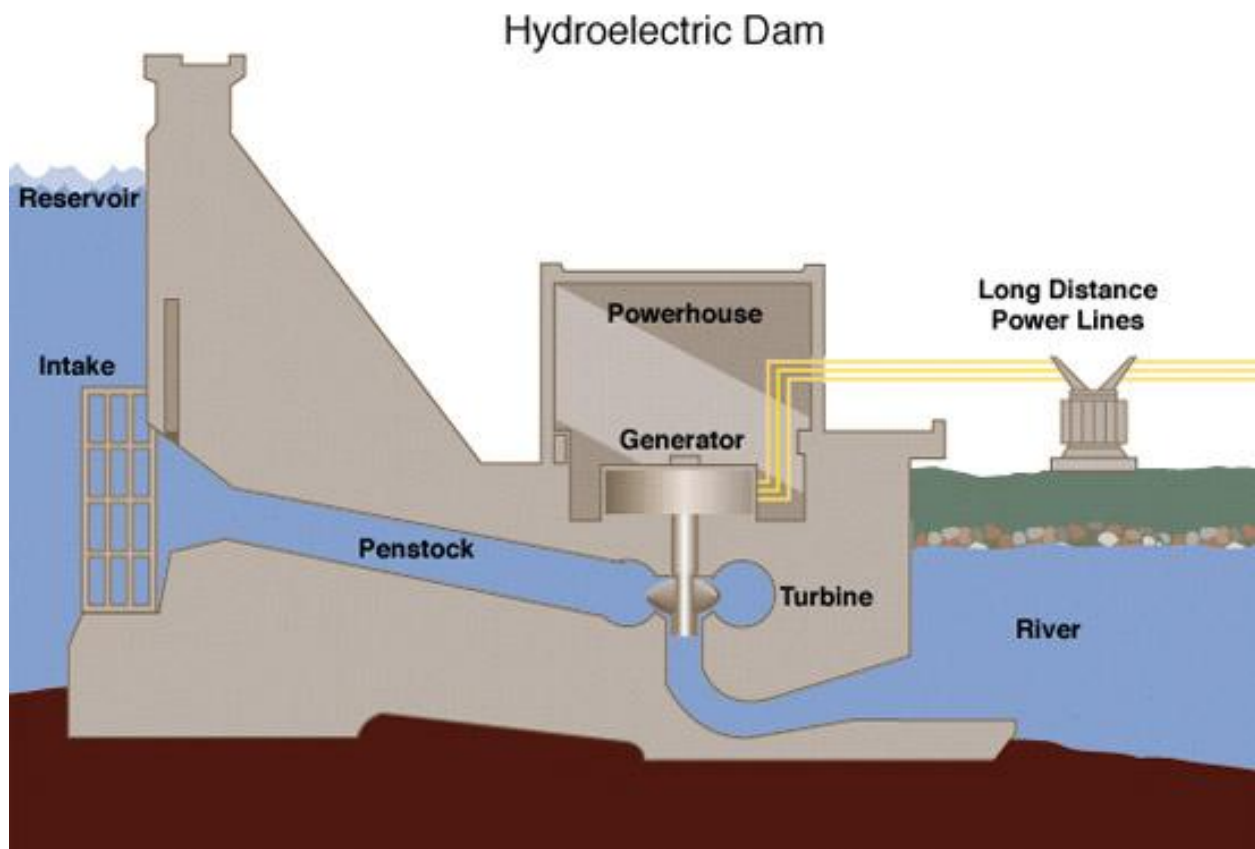
Wind seems to be a very good source of alternative energy. Its biggest setback is its unreliability, but in combination with other, more reliable sources, wind energy should be used extensively to supplement the demand for energy.

Hydroelectric Power

Hydropower is America's leading renewable energy resource. This notable success can be attributed to the fact that out of all the renewable power sources, hydropower the most reliable, efficient, and economical. Furthermore, the concept behind hydroelectric power is fairly simple and has been in use for a significant span of time.

The earliest reference to the use of the energy of falling water is found in the work of the Greek poet Antipater in the 4th century BC. Indeed, the word "hydro" comes from the Greek language meaning "water." Several centuries later, the Romans were the first to utilize the waterwheel. Due to the Romans' powerful influence on Europe through conquest, the waterwheel was soon

commonly found throughout that continent, and by 1800, tens of thousands of waterwheels had been built. These early waterwheels were of course not used for power generation, but mostly for grinding crops. Water energy was first converted into electricity on Sept. 30, 1882 near Appleton, Wisconsin.⁸ By 1980 hydroelectric power accounted for about 25% of global electricity and 5% of total world energy use, which amounted to approximately 2,044 billion kilowatt hours (kW h).



Harvesting energy from water is possible due to the gravitational potential energy stored in water. As water flows from a high potential energy (high ground) to lower potential energy (lower ground), the potential energy difference thereby created can be partially converted into kinetic, and in this case electric, energy through the use of a generator. There are essentially two major designs in use that utilize water to produce electricity: the hydroelectric dam, and the

pumped-storage plant. The waterwheel discussed at the beginning of this paper is currently no longer in use and has been replaced by the far more economical and efficient dam. Both the waterwheel and the dam work on the same general principle, but the dam has the advantage of being more reliable due to the reservoir behind it. The principle is simple: the force of the water being released from the reservoir through the penstock of the dam spins the blades of a turbine. The turbine is connected to the generator that produces electricity. After passing through the turbine, the water reenters the river on the downstream side of the dam. A pumped-storage plant is very similar to the hydroelectric dam, the main difference being that the pumped-storage plant uses two reservoirs, one being considerably higher than the other. The advantage of this design is that during periods of low demand for electricity, such as nights and weekends, energy is stored by reversing the turbines and pumping water from the lower to the upper reservoir. The stored water can later be released to turn the turbines and generate electricity as it flows back into the lower reservoir. Now that the two types of facilities have been discussed, there are also two ways of obtaining the water: dam and run-of-the-river. A dam raises the water level of a stream or river to an elevation needed to create the necessary water pressure. In a run-of-the-river scenario, the water is diverted from its natural path, enters the turbine, and is later returned to the river. Hydroelectric power offers several significant advantages compared to fossil-based power, and even other types of alternative energy. Probably the most important asset of hydroelectric power is its reliability. Furthermore, it creates no pollution, and once the dam is built, even though that process is very expensive, the produced energy is virtually free. A dam has the ability to continuously produce electricity and can adjust to peaks in demand by storing water above the dam and by being able to increase production to full capacity very quickly. Other than the high construction and planning costs, the major drawbacks of large dams are mostly environmental. The dam does not produce harmful emissions as in the case of fossil fuel burning. It does, however, alter the landscape dramatically, producing several severe, even unbearable changes to the habitat of fish and other plants and animals. Building a large dam will, of course, flood a large area of land upstream of the dam, causing problems for the animals that used to live there. It furthermore affects the water quantity and quality downstream of the dam, which in turn affects plants and animals. Blocking the river also disallows certain migration patterns of fish. Finding sites that are suitable for dams is also a challenge. This is one of the reasons why the hydroelectric power production in the U.S. cannot increase by much in the future: most of the

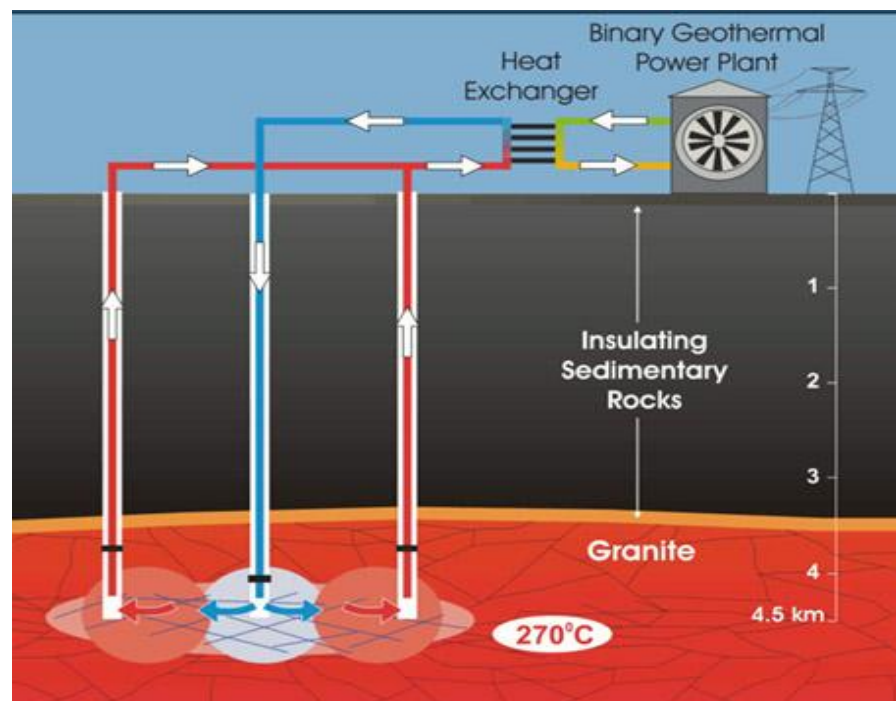
suitable locations have already been utilized. According to the Energy Information Administration, the total amount of electricity produced in the U.S. through hydroelectric means has increased by 6.3% from 2004 to 2005. Even though U.S. construction of dams has peaked and is decreasing, advances in turbine technology maintain a slight growth margin of electricity production. Precipitation however also influences the ability of dams to produce electricity. In this sense, 2005 could have been a year of increased precipitation if compared to 2004. Overall, hydroelectric power seems to be a very good source of alternative energy: one that should be maintained at the maximum level possible. It has the main advantage over all the other forms of alternative energy production in that it is reliable, whereas the other forms of alternative energy are not. The main disadvantage is that hydroelectric energy production in the U.S. is currently being used to its maximum potential, which means that large sums of investment will produce only small increases productivity. Other alternative energy sources are not yet as developed and hence will produce greater advances in productivity with the same or even a smaller input of money. Hydroelectric spending should be maintained at current levels, and more money should be invested in the other sources of alternative energy.

Geothermal

Geothermal energy is one of the only renewable energy sources not dependent on the Sun. Instead, it relies on heat produced under the surface of the Earth. Geothermal energy already has several applications and could potentially provide a significant source of renewable power for the United States. However, it is limited by a multitude of factors revolving around the issues of sustainability and economics. There are two main applications of geothermal energy, which include producing electricity at specialized power plants, and direct-heating, which puts to direct use the temperature of water piped under the earth's surface. Geothermal power plants take on several types of forms, depending on the type of geothermal area from which they extract energy. In any case, the plants depend on steam to power turbines and generate electricity, though the methods of producing steam varies depending on the type of geothermal reservoir.¹² Direct-heating, on the other hand, provides immediate, usable energy. This type of energy can heat individual buildings or entire areas, as in the city of Klamath Falls, Oregon. It can also cool buildings by pumping water underground where the temperature remains relatively stable near 60 degrees Fahrenheit, and then into buildings, where the water absorbs heat, thus helping to air condition the building. The United States also uses direct heating in fish farms, spas, and

greenhouses. Geothermal energy could potentially become a major source of renewable power for the United States. This is because geothermal energy reduces the United States dependence on foreign oil, it's extremely reliable due to the constant source of heat emanating from the earth, and it has almost no negative environmental impact.¹⁵ In 2004, the US produced approximately 2300 MW of electricity, and the Department of Energy estimates that the figure could reach 15000 MW per year within a decade.¹⁶ In the grander scheme, however, geothermal energy accounted for only about 0.34% of total U.S. energy consumption, and 5.56% of renewable energy consumption.¹⁷ But more energy could be extracted using developing technology, which doesn't rely on existing hot water and steam reservoirs. The process involves drilling deep into the surface of the Earth

where temperatures are hot, and then injecting water into cracks of rock, which is heated and then pumped back to the surface.¹⁸ If this "hot dry rock" (HDR) technology proves effective, then more geothermal plants could operate in more locations, since much of the Earth's surface is underlain by hot, dry



rock. Some problems that geothermal energy faces are depletion of both water and heat in geothermal areas. The first problem has been partially addressed by re-injecting water into reservoirs, thus sustaining the plant's ability to operate. However, it has been shown that water re-injection can cause small earthquakes, which raises the question of whether the plants should be liable for the damages caused.¹⁹ In Alameda, California, water reinjection at a geothermal power plant triggered earthquakes of magnitudes up to 3.9 and 3.5 on the Richter scale, which were felt 90 miles away in the community of Middletown.

As of now, there is no government regulation concerning the repayment of damages caused by these earthquakes, though community groups such as the one in Middletown have pressured the plant to compensate homeowners for damages such as cracked chimneys, which can cost about \$10,000 to fix.²⁰ Heat depletion of geothermal areas is more problematic than water depletion in the long run, since it cannot be avoided. It is caused by a natural cooling-off of the earth's crust, and in these cases, plants would become less and less efficient over several decades until they were rendered useless. Other issues facing geothermal power in the United States are building costs and economic competitiveness with other energy sources. Geothermal plants can be expensive, depending on factors such as how deep the wells must be drilled and the temperature of the water or steam. These initial costs of an economically competitive plant can be as high as \$2800 per kW installed capacity, which accounts for about two thirds of total costs for the plant. The plants are economically competitive in the long run however, because their fuel is free, whereas natural gas or coal plants spend up to two thirds of their total operating costs on fuel. Another problem that adds cost to geothermal plants is the problem of connecting to energy grids. This is a critical issue because geothermal plants are built where geothermal resources permit- such as geysers and areas with less-heated water. Over time, however, the plants pay for themselves and all the necessary costs because of low operating costs; namely, the fact that the plants energy is free and always available. The National Commission on Energy Policy believes geothermal energy can cost from 4-6 cents per kWh, which depends on the construction of new geothermal plants, but compares favorably with other renewable energies such as solar power, which costs 20-25 cents per kWh.²² It's also competitive with coal and natural gas, which costs about 4-5 cents per kWh.²³ The projected low cost therefore depends on the availability and exploitation of existing geothermal resource. Because of its reliability, accessibility, low impact on the environment, and potential low cost, geothermal energy is a very attractive source of renewable energy for the United States. Expanding use of geothermal energy depends largely upon the success of the hot dry rock technology and the simultaneous prevention of earthquakes caused by water injection at those plants and water re-injection at other plants. If the HDR technology proves to be viable and safe, geothermal plants can be built in closer proximity to electricity grids, without worrying about geothermal resources like geysers. This would make the plants more cost effective and enable geothermal energy to compete with other energy types.

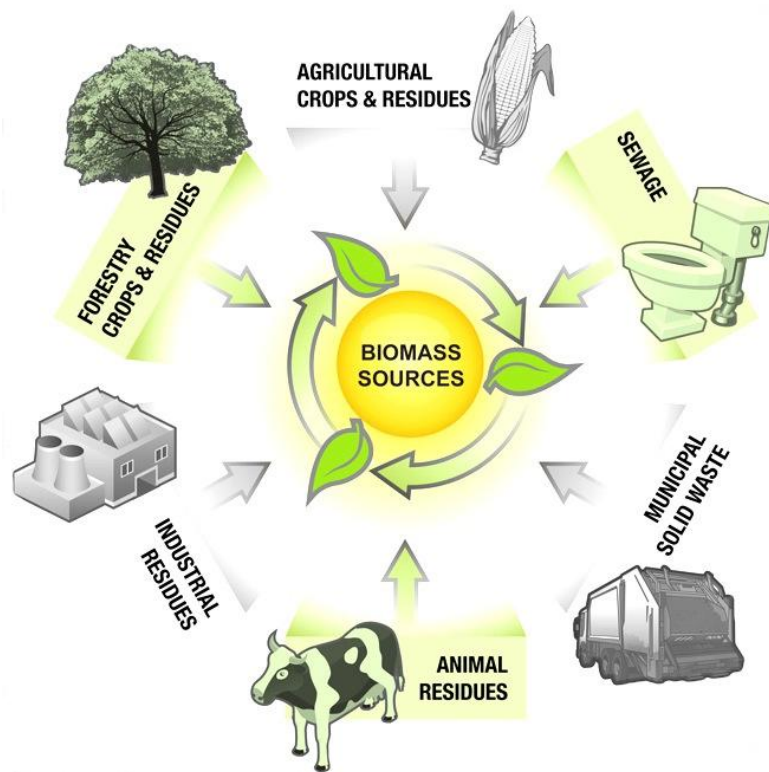
Biomass (Optional : not in the syllabus)

As a pending global energy crisis appears more and more imminent, it is important to consider many different options for new energy sources. Renewable energy sources are ideal because they are more efficient, environmentally friendly and, ultimately, better for consumers. Biomass can be converted into fuels through a number of different processes, including solid fuel combustion, digestion, pyrolysis, and fermentation and catalyzed reactions. Electricity is generated in many places through solid fuel combustion. The majority of America's electricity is fueled by coal combustion. However, many states, especially California, are encouraging companies to use biomass fuels to generate electricity. These products are usually wood matter, vegetation, waste from lumber yards, and the like.²⁵ Power plants burn such fuels to heat a boiler, and the resulting steam powers turbines & generators.²⁶ This process still releases a lot of carbon dioxide and other polluting gases into the environment, but helps eliminate waste efficiently.

Digestion is another process that makes use of existing waste. The term is a misnomer. Digestion is the naturally occurring process of bacteria feeding on decaying matter and making it decompose. It is that which releases gases like methane, hydrogen, carbon monoxide, etc.²⁷ In many landfills, owners are experimenting with set-ups to best collect the gases produced by such bacteria. The standard system includes pipelines running through the waste to collect the gases. Animal feed lots and other facilities are also exploring tapping such resources. A zoo in upstate New York is using their elephant manure to do the same thing. Benefits of this process include the relative lack of impurities in the gases produced and the fact that the synthesis gases (carbon monoxide and hydrogen) can be converted to any kind of hydrocarbon fuel.

A third process, pyrolysis, creates a product much like charcoal, with double the energy density of the original biomass, making the fuel highly transportable and more efficient. Anhydrous pyrolysis heats the biomass at intense temperatures in the absence of oxygen or water. Scientists assume that this is the process that originally produced fossil fuels (under different conditions). Most industrial processes of pyrolysis convert the biomass under pressure and at temperatures above 800° F (430° C). A liquid fuel can also be produced using this process.

The most widely used alternative fuel, ethanol, is created through fermentation of organic materials. Ethanol has a current capacity of 1.8 billion gallons per year, based on starch crops



such as corn. Again, the fuel conversion process takes advantage of a natural process. Microorganisms, especially bacteria and yeasts, ferment starchy, sugary biomass products (like corn), yielding products like ethanol, which can be used as fuels in a variety of applications. Biodiesel is an increasingly popular fuel, especially in the transportation sector. This mono-alkyl ester is formed by combining fuel-grade oil, processed from sources like

vegetable oil, animal fats, algae and even used cooking grease, with an alcohol (like methanol or ethanol), using a catalyst. It shows great promise as both a neat fuel (used alone) and as an additive to petroleum diesel.

Using biomass could be the answer to the energy questions made more imminent by the recent crises that have further threatened our oil supply. The current technologies take advantage of many natural, long-utilized processes in order to create “new” kinds of fuel. Upon further observation, one realizes that these fuels are very basic, using the most readily available energy sources with very simple, standardized processes that greatly reduce pollution and offer hope for the future.

Ethanol

Fuel-quality ethanol is beneficial for car-owners, the economy and the environment. This growing technology is looking to be an immediate part of the solution to the forthcoming energy crisis. Ethanol, also known as ethyl alcohol or grain alcohol, is a colorless, clear liquid. The chemical formula is $\text{CH}_3\text{CH}_2\text{OH}$. Fuel-quality ethanol goes through more processes than do

alcoholic beverages, in order to make it unfit for human consumption and to increase the purity so as to avoid separation when mixed with gasoline. The most common method for making ethanol used in the United States is the dry-mill method. At the beginning of 2005, the 81 ethanol plants in 20 states can produce up to 4.4 billion gallons each year, and the 16 plants under construction are expected to add 750 million gallons of capacity. The dry mill process has advanced to the point at which any cellulosic (plant fiber) biomass can be used to make fuel ethanol (and is now being referred to as the Advanced Bioethanol Technology). The variety of feedstock that can be used today includes corn, barley, wheat, cornstalks, rice straw, sugar cane bagasse, pulpwood, switch grass and even municipal solid waste, offering tremendous opportunities for new jobs and economic growth.

Ethanol is not used by itself to fuel cars. Instead, it's mixed with gasoline. The two most common blends are E10 and E85. The number refers to the percentage of ethanol in the blend. E10 is a blend of ten percent ethanol and ninety percent gasoline. E85, the most mainstream alternative fuel, is eighty-five percent ethanol and fifteen percent gasoline. Using ethanol increases the octane rating and decreases the amount of damaging emissions associated with fuel consumption. It is for this second reason that ethanol use is so strongly recommended and endorsed by state and federal governments. The Clean Air Acts Amendments of 1990 require using reformulated gasoline (RFG) to reduce emissions in major metropolitan areas. RFG blends gasoline with oxygenates, of which ethanol is increasingly popular. Methyl tertiary-butyl ether (MTBE) used to be the most popular, but there are increasing environmental health concerns, regarding seepage, surrounding its use. Oxygenates (compounds with structures similar to that of gasoline, but with the addition of oxygen) dilute the noxious, dangerous gases emitted by gasoline consumption, including nitrogenous oxides, volatile organic compounds and other toxic like carbon monoxide. It is for this reason that the Clean Air Acts Amendments require inclusion of oxygenates like ethanol in the fuel supplies of metropolitan areas, and that the government offers many incentives. These include the Clean Fuel Tax Deduction, taken off the vehicle property tax on new qualified clean fuel vehicles or the conversion of vehicles to run on alternative fuels; the ethanol and biodiesel tax credit, under the American Jobs Creation Act of 2004 (Public Law 108-357); the credit for installation of alternative fueling stations, under the Energy Bill of 2005; the new Flexible Fuel Vehicle labeling requirement, and many more. Approximately one-third of the states offer incentives as well.

Increase in use of ethanol as fuel will benefit farmers economically. The majority of ethanol used today comes from corn, and it is the farmer-owned ethanol plants that are driving the industry's growth. Half of the operating plants are owned by farmers and local investors. The United States Department of Agriculture estimates that the Renewable Fuels Standard would increase the demand for corn for ethanol to 2 billion bushels each year by 2012, almost double the current demand, which would raise net farm income to \$4 billion.

There are drawbacks to using ethanol. The presence of oxygen and smaller molecules means it produces less energy than raw gas, reducing fuel economy by 2 to 3 percent. The octane boost from ethanol is smaller than that of MTBE, and ethanol raises gasoline's volatility, increasing evaporative emissions, all of which are of concern. However, these shortcomings pale in comparison to the health concerns and need to reduce the use of gasoline consumption. There is a reason ethanol blend is required in fuel by Minnesota state law. The environmental and economic benefits make it a desirable alternative. As technology improves, more of the drawbacks will be decreased, and ethanol and other alternative fuels will become mainstream and standard-issue, leading the United States away from our gasoline addiction.

Ocean Energy

Nearly seventy percent of the Earth's surface is covered by oceans, which have the potential to supply humans with an enormous amount of renewable energy. Humans have exploited the vast energy potential of Earth's oceans by taking advantage of wave movement, tides, ocean currents, and ocean thermal energy. The United States, however, has given little or no attention to ocean energy up until this point. This is because of major problems with siting power plants and various economic obstacles. Recent legislation has brightened the future outlook of ocean energy in the United States, but the fledgling technology will take years to realize its potential and account for any significant portion of the United States' consumption of renewable energy. Though the United States does not currently have many power-generating facilities to take advantage of ocean energy, some private and public associations have begun eyeing existing European technologies in hopes of bringing them to American soil or rather, waters. Europe is the world's leader in exploiting ocean energy, due in large part to its location and natural geography. For example, winds blown across the Atlantic from west to east naturally increase the size of waves on Europe's western coast of the west coast of Britain. Larger waves

have greater energy, and therefore more power producing ability. Europe has also led the way in technologies that exploit underwater currents and tides. The United States hopes to learn from these technologies, plus the non-European ocean thermal energy conversion (OTEC) technology, which is feasible only in equatorial waters like those around Hawaii. The city of San Francisco has recently collaborated with a Scottish group called Scotland's Ocean Power on a demonstration project for capturing wave energy, which is one of the four major types of ocean energy.⁴⁹ This technology works on the principle of rolling waves flowing through joints in a large cylindrical pipe, which pushes high pressure oil through hydraulic motors to generate electricity- which is in turn fed to an onshore grid through an underwater cable. Plans are underway to create a "wave farm" off the coast of Britain using these wave energy converters. One square kilometer of ocean interspersed with the devices would produce about 30 MW of electricity, which could power 20,000 homes. About twenty square kilometers could power the city of Edinburgh.⁵⁰ The U.S. has similar ambitions in areas of the Pacific Northwest, like Oregon, Washington, and southern regions of Alaska. In terms of price, the technology in Europe provides electricity at the equivalent of about 9 cents per kW, which is about twice the price of wind power in Europe.⁵¹ Obviously then, the technology has a way to go before it becomes competitive in Europe, and especially the United States, which is further behind. There are numerous other variations of wave-energy systems besides the one mentioned, falling into the categories of onshore, near-shore, or offshore systems; but in any case, the systems manipulate wave motions to power hydraulic pumps or spin turbines, thus generating electricity which is fed via cable to the nearest electricity grid. Americans see another example of ocean energy technology in France, which is home to the world's largest (240 MW) tidal power plant.⁵² Tidal-power plants such as the one in La Rance, France, operate by damming an estuary and generating electricity from water flowing through turbines. There are a number of variations in terms of exactly how electricity is produced, but one popular method is called ebb generation. At high tide, water flows in through openings in the barrage, or dam, spinning turbines to generate electricity. The water is retained behind the barrage until low tide, when it flows out again, once again spinning the turbines and generating electricity. The predictability of tides makes tidal power a reliable energy source, though it can only produce electricity at certain times of day: during high and low tides. Unfortunately, there are only a handful of places in the world where tidal power generation is efficient, and the United States is not home to many. These places

observe a difference of about 5-10 meters between high and low tide, which is ideal and also costs for building these types of projects run high, deterring investors who want quicker returns on their money. Tidal power plants typically have negative impacts on estuarine ecosystems as well, adding a further obstacle to their implementation.

TYPES OF SOLAR CELLS

8.6

1. P-N Homo-Junction solar cells
2. Hetero-Junction solar cells
3. Thin-film solar cells

It can also be broadly divided into crystalline (i.e., monocrystalline and polycrystalline) and amorphous solar cells. *Crystalline solar cells* are normally in the form of solar panels. The output of each cell is between 0.5 and 0.6 volts, and 36 cells are used to produce 20 volts of an open-circuit voltage. This is adequate to charge a 12-volt battery. The most common material used in solar cells is a single-crystal silicon. Its efficiency is about 25% as it is most sensitive to infrared light, and radiation in this region of the electromagnetic spectrum is relatively low in energy. Polycrystalline solar cells are made by a casting process in which molten silicon is poured into a mould and allowed to cool, and then sliced into wafers. The polycrystalline cells are cheaper than single crystal cells, with efficiency less than 20% due to internal resistance at the boundaries of the silicon crystals. Amorphous cells are made by depositing silicon onto a glass substrate from a reactive gas such as silane (SiH_4). Amorphous solar cell can be applied as a thin film to glass or plastic substrates. The advantages of thin-film cells are easier deposition and assembly on inexpensive substrates, the easy mass

production, and the high suitability to large applications. Their efficiencies are presently only about 10% due to significant internal energy losses.

A number of other materials like gallium arsenide, copper indium diselenide and cadmium telluride can also be used to make solar cells. Note that solar cells are sensitive to different wavelengths of light as a function of the materials they are built from. Therefore, some cells are better performers outdoors while others are better performers indoors. Some solar cells have improved energy conversion efficiency by incorporating two or more layers of different materials with different wavelength sensitivities. Top layers are designed to absorb higher energy photons while allowing lower energy photons through to be absorbed by the layers beneath.

8.6.1 P-N Junction Solar Cell

The efficiency of the cell depends on the doping level of the P-region so that electrons generated in this region can diffuse to the junction before they recombine. So a proper match is required between the diffusion length of the electrons L_n , thickness of the P-region and the mean optical penetration depth. A large contact potential V_0 is required to obtain a large photovoltage which is achieved by heavy doping. Long lifetimes are needed for the charge carriers, but heavy doping reduces the lifetime of electrons and holes generated. The series resistance (R_s) of the solar cell should be small so that the power is not lost as heat due to Joule heat in the device itself. This is the major problem in solar cells, especially in the thin region at the surface. So the surface is partially contacted. A series resistance of few ohms can reduce the output voltage to appreciable levels. The resistance of the P-region of the diode of a solar cell is kept low by keeping area of the P-side of the diode large. The N-region is normally thin. This region is contacted at the edge and the current should be along the thin N-region to the contact. This produces large series resistance. The contact fingers distributed over the surface of the N-region will reduce this resistance. Much of the photon flux from the sun has less energy than the cell band gap, so they are not absorbed. For the high-energy photons that are absorbed, the recombination of electron-hole pairs at the surface takes place. This is the reason why the output from a solar cell is limited. A well-prepared Si solar cell is expected to give an efficiency of 10% for solar energy conversion and this gives an output of 100 W/m² of electrical power due to full illumination. This is not cost-effective. Also, the absorbed solar energy increases the temperature of the cell, thereby generating nonequilibrium concentration of charge carriers. The remedy is the use of GaAs and related compound with direct band gaps that can be used up to 100°C. The hetero junctions are very useful for good conversion efficiency and better for operating temperatures with reasonable efficiency. To reduce the surface recombination, the surface of the solar cell is coated with antireflection coating. The maximum output power is achieved by keeping a large-area junction located near the surface of the device. The thickness of the N-region must be less than the diffusion length in the N material.

8.6.2 Hetero Junction Solar Cells

These solar cells use two dissimilar semiconductor materials. The band gap of two materials will be such that $E_{g2} < h\nu < E_{g1}$. Incident photons are absorbed in the material on the top having a band gap E_{g2} . So the photocarriers are generated and diffuse into material with band gap E_{g1} which is beneath the first one. Due to differences in band energies of the two materials, a discontinuity occurs in the energy band diagram of the junction of the two materials. They are very useful for good conversion efficiency and suitable for higher operating temperatures. GaAs-AlGaAs hetero-junction cell is widely used because of good conversion efficiency even at higher temperatures.

8.6.3 Thin-Films Solar Cells

A Thin-Film Solar cell (TFSC), also called a Thin-Film Photovoltaic Cell (TFPV), is a solar cell that is made by depositing one or more thin layers (thin film) of photovoltaic material on a substrate. The thickness range of such a layer is wide and varies from a few nanometres to tens of micrometres.

Many different photovoltaic materials are deposited with various deposition methods on a variety of substrates. Thin-film solar cells are usually categorised according to the photovoltaic material used:

- Amorphous silicon (a-Si) and other thin-film silicon (TF-Si)
- Cadmium Telluride (CdTe)
- Copper indium gallium selenide (CIS or CIGS)
- Dye-sensitised Solar Cell (DSC) and other organic solar cells

Thin-film Silicon A silicon thin-film cell is a thin-film cell that uses amorphous (a-Si or a-Si:H), proto-crystalline silicon, nanocrystalline (nc-Si or nc-Si:H) or black silicon. Thin-film silicon is opposed to wafer silicon (also called bulk or crystalline silicon).

Design and Fabrication The silicon is mainly deposited by Chemical Vapour Deposition (CVD), typically Plasma-Enhanced (PE-CVD), from silane gas and hydrogen gas. Other deposition techniques being investigated include sputtering and hot-wire techniques.

The silicon is deposited on glass, plastic or metal which has been coated with a layer of Transparent Conducting Oxide (TCO).

A P-i-N structure is usually used, as opposed to an N-i-P structure. This is because the mobility of electrons in a-Si:H is roughly 1 or 2 orders of magnitude larger than that of holes, and thus the collection rate of electrons moving from the P- to N-type contact is better than holes moving from P- to N-type contact. Therefore, the P-type layer should be placed at the top where the light intensity is stronger, so that the majority of the charge carriers crossing the junction would be electrons.

Thin-film solar cells are constructed using thin films of semiconducting materials deposited on a suitable substrate like glass, ceramic or metal. Metals and conductors form active substrates whereas insulating substrates are normally passive. Normally, thin-film deposition uses a variety of techniques like thermal evaporation, chemical vapour deposition, molecular beam epitaxy, sputtering, electroplating etc. The films deposited may be polycrystalline or amorphous. Thin-film solar cells are economical but having low efficiencies and stabilities compared to single-crystal solar cells.

Solar-cell arrays are spread over the surface of a satellite to supply power for the electronic equipment in satellites for a long time. This is the advantage over solid state batteries. They can also be attached to the main body of satellites in the form of paddles. They have about 10% efficiency with about 100 W/m² of electrical power under full illumination. To improve the power output, large-area Si cells are fabricated.

ADVANTAGES, ENGINEERING APPLICATIONS AND DISADVANTAGES

8.7

Advantages

- Solar cells are an inexpensive electrical source of energy. Once installed, they can work for years without any expenditure on maintenance.
- Solar cells do not use any fuels. So they are not dangerous.
- They produce enough voltage and current required for operation of equipment in satellites and rockets.
- They neither pollute atmosphere, nor produce noise when they are working, because they do not produce movements and vibrations when they are in operation.
- For small-power applications they can be used to replace chemically operated dry or wet batteries, where their small size is quite convenient.

Engineering Applications The applications found in different fields are rural electrification, water pumping, domestic supply, healthcare, lighting, ocean navigation aids, electric power generation in space, etc.

Disadvantages

- Solar cells are not normally used as optical detectors like photodiodes, because of their very slow response compared to photodiodes.
- To get large output voltage, solar panels occupy large space.
- Solar cells are very delicate. They are brittle, and can be easily broken into small pieces.
- Solar panels must be cleaned from time to time to get better results.
- Solar cells require uninterrupted solar light input. So, during night their operation stops. To get current at night, solar cells should be supported by storage batteries.

Concluding Remark Solar cells, at present, furnish the most important long-duration power supply for satellites and space vehicles. They have also been successfully employed in small-scale terrestrial applications. The solar cell is considered a major candidate for obtaining energy from the sun, with high conversion efficiency, can provide nearly permanent power at low operating cost, and is virtually free of pollution. Recently, research and development of low-cost, flat-panel solar cells, thin-film devices, concentrator systems, and many innovative concepts have increased. In the near future, the costs of small solar-power modular units and solar-power plants will be economically feasible for large-scale use of solar energy.
